Environmental Impacts of Banning Vehicle Advertising*

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Abstract

We develop a structural model of vehicle demand, pricing, and advertising that allows age-specific heterogeneity in preferences, price sensitivities, and advertising responses. The model is estimated using product-level and age-specific data on vehicle sales and advertising in France from 2012 to 2021. We use the estimated model to evaluate firms' pricing and advertising responses to alternative advertising policies. Counterfactual simulations suggest that banning advertising for non-electric and non-hybrid vehicles weighing more than 1,500 kilograms offers the best-targeted policy, generating substantial welfare gains and reductions in CO_2 emissions and vehicle weight comparable to a complete advertising ban.

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1 Introduction

The transportation sector is one of the largest sources of CO_2 emissions and a major contributor to air pollution. Every second, two new vehicles are added to the roads – a number projected by the International Energy Agency (IEA) to rise to more than four by 2030. The environmental impact of cars will largely depend on how effectively we transition to more fuel-efficient vehicles. However, troubling trends suggest that consumer demand is moving in the opposite direction. For instance, the sales of sport utility vehicles (SUVs) have been increasing steadily each year.[1] A 2022 report by the IEA notes that the world's fleet of SUVs collectively emitted almost one billion metric tons of CO_2 – an amount equivalent to the combined national emissions of the United Kingdom and Germany in the same year. This trend has now extended to electric vehicles (EVs): in 2022, sales of electric SUVs surpassed those of other types of electric cars. In France, both the SUV share of all new-vehicle sales and the SUV share of electric/hybrid sales doubled from 2012 to 2021.[2] Heavier vehicles, whether conventional or electric, require more energy to move, meaning that until global electricity generation becomes carbon-free, heavier EVs will produce more emissions.

The automotive industry is among the top spenders on advertising worldwide, and was the second-biggest sector in the French advertising market until 2019. Automotive brands are particularly dependent on television advertising. In France, the share of automotive advertising spending on TV has been growing continuously since 2012.[3] Evolving consumer needs and technological changes over the past few years have been accompanied by a shift in the industry's marketing focus. Brands have increasingly directed their advertising budgets toward their SUV ranges and their "green" (i.e., electric or hybrid) models.[4]

To date, policymakers have primarily focused on promoting technological solutions to tackle climate change, such as green innovation and emissions performance standards. However, effectively reaching the net-zero emissions target also requires shifts in consumer behavior. Governments may need to go further than imposing vehicle taxes or subsidizing electric and hybrid car purchases. Today, automotive manufacturers continue to heavily promote their most polluting products, despite the enormous amounts of CO_2 they release into the atmosphere, putting the planet at risk. The European Citizens' Initiative to ban fossil fuel advertising and sponsorships, launched by Greenpeace in 2021, collected 353,103 signatures across Europe in just one year. Following this campaign, national bans on fossil fuel advertising have been proposed in countries such as Spain, the Netherlands, and Sweden, while France has already incorporated such a ban into its climate law. France

[1] On average, an SUV consumes about 20 percent more fuel than a medium-sized car to travel the same distance. Since fuel use translates directly into CO_2 emissions, SUVs also release approximately 20 percent more carbon per mile driven.

[2] We show these trends in Figures A1 and A2 in the Online Appendix.

[3] See Figure A3 in the Online Appendix.

[4] Figures A4 and A5 in the Online Appendix respectively show the growth in the shares of SUV and electric/hybrid vehicle advertising expenditure on French TV. Figure A6 shows the significant increase in advertising spending for electric and hybrid SUVs since 2019.

will ban advertising for the most polluting vehicles, including many SUVs, starting in 2028.

In this paper, we evaluate the environmental and welfare effects of vehicle advertising bans. We develop a structural model of vehicle demand, pricing, and advertising and use it to simulate market equilibrium outcomes under a range of advertising restrictions, including a complete advertising ban and bans targeting the most polluting vehicles. The framework captures heterogeneity in vehicle demand across consumer age groups and allows consumers' choices to depend on their advertising exposure. In addition, we allow advertising for new vehicles to influence demand for second-hand vehicles. The model characterizes equilibrium vehicle prices and age-specific demand under alternative advertising policies. Under partial advertising bans, it also determines how brands optimally reallocate advertising across vehicles.

We model the demand for new vehicles using a random-coefficients discrete choice framework, allowing for heterogeneity in advertising exposure and vehicle-price sensitivity across consumer age groups. To account for substitution toward used cars, we additionally model the demand for second-hand vehicles using a simple logit framework. We estimate the model using longitudinal data on vehicle sales, advertising, and television viewership in France between 2012 and 2021. We observe quarterly sales of different car models at the age-group level. Our sales data include vehicle characteristics (such as horsepower, weight, body style, and CO_2 emission levels), as well as vehicle retail prices. We couple this with data on product-level television advertising and detailed TV viewership measurements. Our advertising data are recorded at the occurrence level, where an occurrence is the placement of an ad for a specific product (a brand and vehicle-model pair) on a given channel on a given day and time. The viewership measurements include daily estimates of the number of viewers of each channel by age group and time block (e.g., 06:00-08:59, 09:00-11:59, etc.). These data enable us to construct age-group-specific exposures to each advertisement. Identification of advertising effects relies on rich variation in advertising exposure across vehicle models, consumer age groups, and time. To address the endogeneity arising from advertisers targeting consumers based on predictable demand shocks, we control for a rich set of interactions between age groups, vehicle characteristics, product fixed effects, and time fixed effects. In addition, we allow unobserved demand shocks to be serially correlated, thereby accounting for demand persistence that may induce further endogeneity in advertising decisions.

On the supply side, vehicle pricing and advertising decisions are made sequentially. Profit-maximizing vehicle prices depend on consumers' current advertising exposure across different vehicles, while today's advertising decisions influence the pricing equilibrium in the subsequent period. Car manufacturers outsource the purchase of advertising slots to specialized agencies, which acquire advertising inventory on behalf of the brand subject to a given budget. The brand then decides how to allocate the purchased advertising slots across its vehicle models. A vehicle brand typically offers several models for sale but concentrates its advertising on one or a few models at a given time. When making these allocation decisions, the brand seeks to maximize expected vehicle-sales revenues in the subsequent period. Given the large number of brands and vehicle-models in the

market, solving the resulting game, that is, determining the equilibrium advertising allocations, prices, and demand, is computationally challenging and may give rise to multiple equilibria. To address this issue, we develop an efficient method for constructing approximate bounds on equilibrium outcomes that combines machine-learning techniques with an equilibrium-search procedure.

We consider a range of advertising restrictions in our counterfactual analysis, including (i) a complete advertising ban, (ii) a ban on SUV advertising, (iii) bans on advertising for high-CO_2-emission vehicles, (iv) bans on advertising for high-weight vehicles, and (v) bans on advertising for non-electric and non-hybrid high-weight vehicles. For the high-emission and high-weight criteria, we consider a wide range of CO_2 and weight thresholds to define the set of restricted vehicles. Evaluating a complete advertising ban only requires solving for the new pricing equilibrium. However, assessing the effects of targeted advertising bans on polluting vehicles requires solving for both the pricing and advertising equilibria, as brands respond by reallocating advertising toward unrestricted vehicles.

Findings: Our demand estimates indicate a significant positive effect of advertising on vehicle sales. The unweighted average own-advertising elasticity of demand is approximately 0.16, which is in line with existing estimates in the literature. We also find evidence of a positive spillover from advertising for new vehicles to demand in the second-hand market, although the corresponding elasticities are roughly an order of magnitude smaller than the own-advertising elasticities for new vehicles.

Our counterfactual analysis identifies a ban on advertising non-electric and non-hybrid vehicles weighing more than 1,500 kilograms as the most promising targeted intervention. The policy generates the largest welfare gains among the policies considered and achieves reductions in total CO_2 emissions and vehicle weight that are comparable to those of a complete advertising ban.

Related Literature and Contributions: Our paper contributes to the literature on environmental policies in the automobile market. Previous work has primarily focused on fuel economy standards (Goldberg, 1998; Jacobsen, 2013; Grigolon et al., 2018; Davis and Knittel, 2019; Levinson, 2019; Reynaert, 2020), feebate policies (Adamou et al., 2014; D'Haultfœuille et al., 2016; Durrmeyer and Samano, 2018; Durrmeyer, 2022) and electric vehicle subsidies (Remmy, 2026; Fournel, 2024; Barwick et al., 2024). To the best of our knowledge, this is the first academic paper that considers advertising bans on polluting vehicles as an environmental policy tool.

Accordingly, our paper contributes to the empirical literature examining the effects of advertising bans. Dubois et al. (2018) consider the effect of banning advertising in the junk food market. Tuchman (2019) investigates the likely consequences of banning advertising for e-cigarettes. Sinkinson and Starc (2019) measure the potential effects of banning direct-to-consumer advertising by drug manufacturers. Abi-Rafeh et al. (2026) analyze the dynamic interaction between advertising

and sin taxes. We are the first to extend this literature to the automotive market.[5] Beyond a complete advertising ban, we simulate the potential effects of partial bans and develop a framework that allows advertisers to optimally allocate their purchased advertising slots.

From a methodological perspective, we contribute to the large literature on the equilibrium demand and pricing of vehicles (e.g., Berry et al. 1995, henceforth BLP). We extend this work by incorporating the impact of advertising exposure on consumers' vehicle choices. The advertising literature has primarily focused on the effects of *brand-level* advertising in other markets (e.g., Honka et al. 2017, Shapiro 2018, Shapiro et al. 2021, and Abi-Rafeh et al. 2026), whereas advertising in the vehicle market is largely conducted at the *product-level*. Advertising agencies purchase advertising inventory on behalf of vehicle brands subject to a given budget, while brands decide how to allocate the acquired advertising slots across vehicle-models. We formalize this advertising-allocation decision and incorporate it into an equilibrium model of vehicle demand, pricing, and advertising. A key methodological challenge is that the market features many brands and vehicle-models, resulting in a large-scale game with potentially multiple equilibria. We address this challenge by developing an efficient procedure that combines machine-learning tools with an equilibrium-search algorithm to construct approximate bounds on equilibrium outcomes.

2 Data

2.1 New-Vehicle Sales and Characteristics

The first dataset we use is obtained from AAA Data and contains information on sales, prices, and product characteristics for all new passenger cars sold in France between 2012Q1 and 2021Q4. Our sales data are at the vehicle-model and age-group level at a quarterly frequency. A vehicle-model j is defined as a brand and model-name pair (e.g., Volkswagen Golf). We observe sales by four age groups: 0-24, 25-34, 35-49 and 50 and above. We directly observe the main characteristics of each vehicle such as horsepower, weight, body style (enclosed, convertible, or station wagon), engine type (including gasoline, diesel, LPG/gasoline, superethanol, electric, and six variants of hybrid engines), average fuel cost, as well as the vehicle's CO_2 emission level. The CO_2 emission level is the number of grams of CO_2 emitted per kilometer measured in standard driving cycle tests.

Sales are defined as new vehicle registrations. Prices are suggested retail prices including VAT. We obtain net vehicle prices after adjusting these suggested retail prices to the French feebate system. The feebate is a financial rebate to consumers who buy less-polluting vehicles, and an extra fee to consumers who buy more-polluting vehicles. The exact amount of rebate or fee depends on the vehicle's CO_2 emission level. There have also been several changes to the feebate program during our sample period, which contribute to variation in net vehicle prices over time. The vehicle's

[5]Relatedly, Murry (2017) focuses on how advertising decisions can impact the contracting between car manufacturers and their dealers and Barroso and Llobet (2012) focus on the informative role of vehicle advertising, building on the framework of Goeree (2008).

fuel cost is obtained by multiplying its average fuel consumption (in liters per 100km) by the fuel price, following D'Haultfœuille et al. (2016) and Grigolon et al. (2018). Finally, monetary variables (net vehicle prices and fuel cost) are deflated to be expressed in constant 2018 euros.

2.2 Advertising Volume and Expenditure

The second dataset we use includes product-level television advertising data over 2012-2021, obtained from Kantar Media. The advertising information is recorded at the occurrence level, where an occurrence is the placement of an ad for a specific product (a vehicle-model) on a given channel, date and time. We observe the length of each ad, together with the cost of placing it. All advertising aired on national broadcast and cable TV channels is contained in the data. We observe rich variation in the ad placements – in terms of channels and airtime – which results in differentiated exposure for consumers for different vehicle-models.

2.3 TV Viewership

The third dataset, obtained from Médiamétrie, contains daily measurements of the number of viewers of each channel by five age groups (0-14, 15-24, 25-34, 35-49, 50+) and different time blocks (e.g., 06:00-08:59, 09:00-11:59, etc.) for each day over 2012-2021. As documented in Zhang (2024), these age groups and time blocks capture the relevant variation in TV viewership during our sample period. We match the TV viewership data with the ad occurrences data to obtain an estimate of the number of impressions for each ad for each age group (0-14, 15-24, 25-34, 35-49, 50+).

2.4 Merging the Ad Impression and Vehicle Sales Data

Merging the advertising and sales datasets is relatively straightforward as we directly observe the precise brand and model name of each vehicle in both. The main challenge that we have to overcome is that the two datasets have a different coding standard for the same vehicle-model. We first perform approximate matching of vehicle-model names, then inspect and manually fill in the unmatched vehicle-models to build the final combined dataset. Many vehicle-models that register positive sales are not advertised in the same year. On the other hand, brands often start to advertise a new vehicle-model before we observe any positive sales for the model. In our modeling, we consider a vehicle-model j to be available for purchase if we observe positive sales for it during the year. We do, however, take into account the advertisements placed for a vehicle-model before it enters the market in our measurement of the consumers' advertising exposure states, which we detail in Section 3.3.

2.5 Additional Data

We obtain data on the total French population over 2012-2021 split by the same four age groups from the French National Survey Institute (INSEE). These data are used to compute the consumers' advertising exposure states and vehicle market shares, which we detail below.

To complement the primary dataset on new vehicle sales, we also obtain from AAA Data vehicle model-level characteristics and age group-specific quarterly sales of second-hand vehicles in 2018. Price information, however, is unavailable for second-hand vehicle transactions. We use the second-hand sales data in our counterfactual analysis to assess how various advertising bans would affect the overall emissions and weight of the vehicle fleet.

Finally, we obtain annual data from Kantar Media on industry-level total advertising spending on media other than TV from 2012 to 2021, as well as all vehicle models advertised each year on these non-TV media. From the same source, we collected annual information on the advertising agencies employed by different vehicle brands. The spending data is available only at an aggregate industry level, covering all car manufacturers combined rather than individual firms. These data confirm that TV has remained the dominant advertising platform throughout the sample period (see Figure A3).

The other media include radio, newspapers, cinema, outdoor, internet display, digital video, paid search, and paid social media. Although we only observe aggregate advertising data for these media rather than separate data for each channel, an important observation is that the same vehicle models are consistently advertised across all platforms, with an exact yearly match between TV and non-TV media. Notably, SUVs and electric/hybrid vehicles are increasingly featured across all media. We therefore focus on TV advertising to exploit the available TV viewership data and construct measures of consumers' advertising exposure in what follows. Accordingly, our estimates should be interpreted as capturing the effect of TV advertising exposure which, given the close alignment of advertising campaigns across media, is also informative about the broader effect of advertising exposure.

3 Descriptive Analysis

3.1 The New-Vehicle Market in France

There are between 19 and 22 firms active in the French new-vehicle market in each year of our data.[6] Some firms (e.g., Honda, Tesla) own only one brand, but many others (e.g., Volkswagen, Renault) own multiple brands. In total, we observe between 37 and 41 brands in the market each year. The majority of these offer various vehicle-models for sale. At most, we observe a brand

[6]In total, we observe 27 unique firms in our data but there have been two mergers and acquisitions and several entries and exits of smaller firms during the sample period. Out of these 27 firms, 16 are multinational giants (e.g., Volkswagen, PSA, Toyota) that are always active on the market.

selling 34 different vehicle-models in a year.

We use the standard international car classification system (ISO 3833-1977) to group vehicle-models into six segments. These are super-mini vehicles (segment A/B); small family vehicles (segment C); large family vehicles (segment D); executive-, luxury- and super-cars, also called upper class (segment E/F/S); minivans, also called multi-purpose vehicles (segment M); and SUVs (segment J). Most brands sell vehicle-models in more than three segments, but there are also eight brands that are specialized in one specific segment. For instance, the brand Jeep specializes in SUVs, while the brand Alpine specializes in the upper class segment.

After excluding vehicle-models with extremely low sales (where the combined share of vehicles we exclude is below one percent of total sales) and very high prices, our sample includes between 162 and 197 vehicles per quarter. Model-level entry and exit is very common in the new-vehicle market. A notable trend in our sample is the increase in demand for SUVs at the expense of demand for vehicles in other market segments.[7] Furthermore, electric and hybrid vehicles have seen a surge in popularity since 2020.[8]

3.2 Advertising Slots Purchases and Allocations

As discussed above in Section 3.1, a vehicle manufacturer (e.g., Volkswagen) may own multiple brands, each offering several vehicle models. However, a manufacturer's advertising budget is not divided equally across its brands. In practice, firms typically allocate most—if not all—of the budget to a single brand, usually the one they have historically prioritized in advertising.

Vehicle brands outsource the purchase of advertising slots to agencies, which negotiate with TV channels on their behalf. The data show that each brand works with a single agency within a given year. When multiple brands under the same firm engage in advertising, it is common for each brand to contract a different agency.

Automotive advertising is primarily conducted at the vehicle-model level rather than the brand level. Brands allocate their purchased advertising slots across individual models and, in practice, rarely advertise all models simultaneously. Instead, they typically promote at most one model per segment in a given quarter. This pattern holds for 93 percent of brand–segment–quarter observations. Of the remaining 7 percent, 6.5 percent involve promoting two models, whereas only a very small fraction involve three or four models.

Figure A8 in the Online Appendix plots quarterly brand-level advertising expenditures for the 15 brands that spend the most on advertising in the super-mini, small-family, and SUV segments. Each color represents one brand. The figure shows that many brands advertise simultaneously within the same segment and quarter, indicating intense competition to promote their models across all three segments.[9] Notably, no single brand consistently dominates advertising spending

[7] We plot this trend in Figure A1 in the Online Appendix.

[8] We plot this trend in Figure A7 in the Online Appendix.

[9] In practice, more brands are active in advertising than can be distinctly represented in the figure.

across all segments, although the three long-established domestic brands (Citroën, Peugeot and Renault) tend to spend more than most others.

3.3 Individual Advertising Exposure

Denote the set of age groups by $\mathcal{D} = \{0\text{-}24, 25\text{-}34, 35\text{-}49, 50+\}$. We define the total exposure to advertising for an individual in age group $d \in \mathcal{D}$ for vehicle j in year-quarter t as:

$$a_{jt}^d = \sum_{s \in \mathcal{S}_t} w_{st}^d T_{jst} \tag{1}$$

Here, s indexes advertising slots across channels and $\mathcal{S}_t$ is the set of slots in quarter t. The variable T_{jst} is the duration in minutes vehicle-model j was advertised in slot s in quarter t. The variable w_{st}^d is the probability that someone from age group d viewed the adverting in slot s. We measure this probability using $w_{st}^d = \frac{v_{st}^d}{pop_t^d}$, where v_{st}^d is the number of viewers from age group d viewing the channel during the time block containing slot s, and pop_t^d is the total number of individuals in age group d in the French population in quarter t. Together, $w_{st}^d T_{jst}$ is the expected exposure to a single ad. We sum over slots to obtain the total expected exposure a_{jt}^d over the whole quarter t.

We assume a dynamic effect of advertising on demand such that the advertising exposure state for an individual is equal to the discounted sum of current (up until the quarter before we observe the purchase being registered) and past advertising exposure, as in Erdem et al. (2008), Dubois et al. (2018), and Shapiro et al. (2021):[10]

$$A_{jt}^d = \sum_{n=\bar{t}}^{\bar{t}} \delta^{n-1} a_{j,t-n}^d \tag{2}$$

In estimation, we use $\bar{t} = 1$, $\bar{t} = 4$, and $\delta = 0.5$.[11] Figures A11 and A12 in the Online Appendix plot the advertising exposure by age group for some popular models in different segments. Figures A13 and A14 in the Online Appendix compare, respectively, the flows and stocks of exposure for one brand's (Peugeot) most advertised models in each segment. Exposure to advertising for various

[10] We also considered alternative functional forms, used in Dubé et al. (2005), Shapiro (2018) and Dubois and Majewska (2022), which define the stock of advertising as $A_{jt}^d = \sum_{n=\bar{t}}^{\bar{t}} \delta^{n-1} \log(1 + a_{j,t-n}^d)$ or $A_{jt}^d = \sum_{n=\bar{t}}^{\bar{t}} \delta^{n-1} \sinh^{-1}(a_{j,t-n}^d)$. We refer to the former as the logarithm-plus-one specification and the latter as the inverse-hyperbolic-sine specification. Figure A15 in the Online Appendix shows that the estimated elasticities are similar across specifications, indicating that our results are robust to alternative definitions of advertising stock.

[11] We use four lags and a decay parameter of 0.5 following Dubois and Majewska (2022) who also use quarterly data. We stop at four lags to avoid losing more time periods in estimation and because beyond four lags advertising exposure has negligible effects. Using $\delta = 0.5$ is also consistent with Dubois et al. (2018) and Shapiro et al. (2021) who use a decay parameter of 0.9 with biweekly data: based on an average of 6.52 fortnights per quarter, $0.9^{6.52} \approx 0.503$. We additionally experiment with alternative carryover rates, varying δ from 0.1 to 0.9. Figure A15 in the Online Appendix reports the estimated elasticities. We find that the estimates are largely insensitive to the choice of carryover rate.

vehicles is differentiated across age groups. Overall, individuals aged 35 and above are more exposed to car advertising than younger age groups. Exposure to advertising for a specific vehicle over time generally follows the same pattern of variation across different age groups, but there are also exceptions. We note, for instance, that the advertising exposure stock for Dacia's "Duster" model increased between 2018 and 2020 for those aged 50 and above, but decreased for individuals aged 34 and under (see Figure A11). Similarly, the advertising exposure stock for Renault's "Captur" model increased between 2018 and 2021 for individuals aged 50 and above, but has decreased for younger age groups (see Figure A12). We also note that advertising intensity varies substantially over time across vehicles and segments. This rich variation is valuable for identifying advertising effects in the estimation.

3.4 Correlation between Advertising and Sales

Overall, the most advertised (and also the best-selling) vehicle-models are in the super-mini, small family and SUV segments, and the share of SUVs in both the advertising budget and sales is growing every year. Only a limited number of brands advertise their models in the upper class segment. The most-promoted model does not necessarily register the highest sales within a brand-segment. In practice, brands often (although not exclusively) promote their new models, but their existing models may sell better even without any advertising during the year. The correlation between sales and advertising flows is 0.36 at the brand–model–year–quarter level, whereas the correlation between sales and advertising stocks is stronger, at 0.44. At the firm level, the brand with the highest sales is typically the one the firm has historically prioritized in its advertising.

Both advertising and sales in the automobile industry move in line with the business cycle of the overall economy. The market grew steadily from 2015 to 2019, after the market recovered from the crisis of 2008. It fell sharply again in 2020 due to the COVID-19 pandemic and the subsequent energy crisis.[12]

3.5 Advertising Exposure Effects

Estimating the effects of advertising on sales is generally challenging. In our setting, we observe rich variation in exposure to advertising at the vehicle-model level across consumer age groups and over time. Some of this variation may reflect brands' attempts to target age groups and/or time periods in which demand is particularly susceptible to advertising. However, because segmented TV advertising was not permitted during our sample period, targeting could only be achieved indirectly through the choice of channels and advertising times. The variation in our advertising exposure measure defined in equation (1) is driven by the age-specific estimates of the viewing probability w^d_{st}, coupled with vehicle-models assigned to each of the advertising slots T_{jst}. A

[12]We plot the total advertising expenditure and total sales over time in Figures A9 and A10 in the Online Appendix.

threat to the identification of the effect of advertising from using this form of variation is that brands successfully target individuals in a particular age group based on certain characteristics of a vehicle. In our regressions, we control for such targeting by including interactions between age and key vehicle characteristics (body style, horsepower, fuel cost, and weight), as well as age–product and age–quarter–year fixed effects. We further include segment–quarter–year fixed effects to account for periodic promotion of vehicles within a specific market-segment (e.g., SUV). We thus exploit the residual variation, which is age-product-year-quarter specific, to estimate the causal impact of advertising on vehicle sales.[13]

Endogeneity of the advertising variable may arise if brands were able to perform sophisticated high-frequency targeting according to transient demand shocks that are unobservable (to econometricians) and specific to each age-product-year-quarter. The institutional features of the ad-buying process in the French TV market makes such precise targeting unlikely. First, all TV advertising slots are sold in an upfront market well in advance of the ads being aired. There is not even a scatter market, like in the US, where last-minute purchases of individual advertising slots are possible. Second, car manufacturers purchase advertising slots through agencies. Both advertising agencies and TV channels may face competing demand from different advertisers for the same audience reach. The allocation of advertising slots under such circumstances is unclear; however, timely targeting at the age-product level is difficult for any advertiser. More importantly, data indicate that advertising for new vehicle models begins before any sales are observed, confirming that campaigns are planned prior to the realization of age–product–quarter–year-specific demand shocks.[14] Furthermore, television program audiences are generally not age-restricted, limiting the ability of advertisers, their agencies, and TV channels to control the age composition of a campaign.

Table 1 presents reduced-form estimates of the demand response to advertising. The results indicate a positive and statistically significant effect of advertising exposure on vehicle sales. Consistent with the intuition that brands target age groups and periods with higher expected responsiveness to advertising, the estimated advertising effect is attenuated when allowing for interactions between age-group indicators and key vehicle characteristics, as well as product and year–quarter fixed effects. In other words, our rich set of fixed effects helps address endogeneity concerns arising from targeted advertising based on predictable demand shocks. The standard errors reported in parentheses are clustered at the age–year-quarter level, which provides a more conservative inference. We note that standard errors increase when clustering at the age–product level rather than at the age–year–quarter level, suggesting the presence of serial correlation in the error terms. Because advertising has lagged effects, it is not surprising that demand shocks exhibit serial correlation. We

[13] A linear regression of the advertising exposure stock A^d_{jt} on the interactions between age and the vehicle characteristics, age-year-quarter fixed effects, and vehicle fixed effects has an R^2 of 0.428. This indicates that, conditional on our controls for targeting, there is still substantial residual variation in the advertising exposure stock A^d_{jt}.

[14] Endogeneity of the advertising variable may still arise due to persistence in demand shocks, for instance when brands form expectations about future demand based on past realizations. We address this issue explicitly by modeling serial correlation in the demand shocks within the structural demand estimation.

Table 1: Advertising Exposure and Vehicle Sales: Reduced-Form Evidence

Dependent variable: Sales	(1)	(2)	(3)
Advertising exposure stock	4.452	3.649	1.864
	(0.406)	(0.384)	(0.264)
Price	−2.996	−2.931	−2.805
	(1.127)	(1.039)	(0.787)
Product characteristics controls	Yes	No	No
Product characteristics controls × Age	*No*	*Yes*	*Yes*
Product fixed effects	Yes	Yes	No
Product fixed effects × Age	*No*	*No*	*Yes*
Year-quarter fixed effects × Age	No	No	Yes
Segment-year-quarter fixed effects	Yes	Yes	Yes
Differentiation IVs for price	Yes	Yes	Yes
Kleibergen-Paap F-statistic	78.69	78.65	80.23
Number of observations	26384	26384	26384

Standard errors clustered at the age–year-quarter level are reported in parentheses. Price is measured in €100 and is instrumented using Differentiation IVs in the spirit of Gandhi and Houde (2019). For each period t, we calculate the distance in characteristic ℓ, $d^l_{jj',t}$, between every pair of vehicles j and j' and use it to construct a closeness weight, $1/(1 + d^l_{jj',t})$, which we sum over rival brands' products. The validity of these instruments is discussed in Section 5.

formally account for this persistence in the structural demand estimation below.

4 Equilibrium Model

The descriptive analysis suggests that advertising has a positive and significant impact on vehicle sales, indicating that consumers' vehicle choices can be influenced by policy interventions targeting advertising. We now introduce an equilibrium model of vehicle demand, pricing, and advertising, which serves as the basis for simulating the counterfactual effects of various advertising bans in Section 7.

We model demand for vehicles using a random-coefficients discrete choice framework, allowing for heterogeneity in advertising exposure and vehicle-price sensitivity across consumer age groups, as well as serial correlation in demand shocks. On the supply side, car manufacturers set vehicle prices and advertising sequentially. Profit-maximizing retail prices depend on consumers' current advertising exposure across different vehicles, while today's advertising decisions influence the pricing equilibrium in the subsequent period. To operationalize these choices, advertising decisions follow two steps: first, advertising slots are purchased; second, the purchased slots are allocated

across vehicles. Car manufacturers outsource the purchase of advertising slots to agencies. The agencies simultaneously choose which slots to buy for the brands they represent, aiming to maximize each brand's total advertising exposure subject to its budget constraints. Afterward, the different vehicle brands simultaneously allocate their purchased slots across individual vehicles. Because advertising occurs at the vehicle-model level and brands typically advertise only a few models simultaneously, the advertising-slot allocation game is particularly important for our counterfactual analysis. Partial advertising bans targeting different sets of polluting vehicles primarily operate by restricting the set of models eligible for advertising, thereby altering the feasible allocation choices available to brands.

4.1 Demand

Our demand model builds on Berry et al. (1995) and the subsequent literature, allowing for rich substitution patterns through both discrete sources of market segmentation and continuous product characteristics. An additional key feature of our model is that it incorporates the impact of advertising exposure on consumers' vehicle choices.

Specifically, time periods (year-quarters) are indexed by t and the set of time periods is $\mathcal{T}$. In each period t, there are M_t^d potential consumers in age group d. Each consumer i may either purchase a new vehicle j from the set of differentiated vehicles on offer $\mathcal{J}_t$ or choose the outside good $j = 0$.[15] Vehicle j is produced by brand b, which is owned by firm f.

Let $\mathcal{J}_t$ be the set of new vehicles offered in period t and $\mathcal{J} = \bigcup_{t\in\mathcal{T}} \mathcal{J}_t$ be the full set of products available in the new-vehicle market over the sample period. We allow the indirect utility for a consumer from purchasing a vehicle j to depend on several factors, including the vehicle's price p_{jt}, observed characteristics (horsepower, weight, body style, fuel cost, segment) $\boldsymbol{x}_{jt}$, and the consumer's advertising exposure stock for that vehicle A_{jt}^d. Formally, we specify the conditional indirect utility of consumer i in age group d from purchasing vehicle model j in period t as follows:

$$u_{ijt}^d = \alpha_i^d p_{jt} + \beta^d A_{jt}^d + \boldsymbol{x}_{jt}'\boldsymbol{\gamma}^d + \vartheta_j^d + \varsigma_t^d + \nu_{gt} + \xi_{jt}^d + \varepsilon_{ijt}^d \tag{3}$$

Here, α_i^d is the consumer's sensitivity to price p_{jt}, which is allowed to be heterogeneous across consumers and is modeled as $\alpha_i^d = \alpha^d + \sigma v_i$, where $v_i \sim \mathcal{N}(0,1)$. The parameter σ is the standard deviation of price sensitivities and the parameters α^d are the mean price sensitivities that are allowed to vary by age group. The term $\beta^d A_{jt}^d$ captures the effect of vehicle-model–level advertising, which we allow to vary across age groups.[16,17] Through $\boldsymbol{\gamma}^d$ we allow for age-group-specific preferences for

[15] The outside good includes both used cars and alternative means of transportation (i.e. public transportation, cycling, etc.).

[16] In our modeling, we explored a specification in which the parameters of the advertising effects were also random, but the estimated standard deviations of the advertising effects were close to 0.

[17] We additionally considered including a second advertising variable capturing brand-level spillovers, but found these effects to be small and not economically meaningful.

observable vehicle characteristics $\boldsymbol{x}_{jt}$. We further include age-group-specific product fixed effects ϑ_j^d to capture differential preferences for individual vehicle models across age groups, and age-group-specific year-quarter fixed effects ς_t^d to control for nationwide fluctuations in overall vehicle demand that may also vary by age group. Given that our sample spans ten years, we additionally incorporate segment-year-quarter fixed effects ν_{gt} to account for changes in the popularity of different vehicle segments (e.g., SUVs) over time. As detailed in Section 3.1, the vehicle market is segmented and each vehicle belongs to a segment $g \in \mathcal{G}$ (mini, large family, SUV, etc). The term ξ_{jt}^d captures the demand shocks from age group d in year-quarter t for product j that is observable to consumers and firms but not to the econometrician. Our reduced-form analysis in Section 3.5 suggests serial correlation in these shocks; therefore, we model this term as evolving according to a first-order autoregressive process with drift (AR(1)): $\xi_{jt}^d = \lambda_0 + \lambda_1 \xi_{j,t-1}^d + \tilde{\xi}_{jt}^d$. Finally, ε_{ijt}^d is consumer i's individual-specific valuation for vehicle j, which is assumed to be identically and independently distributed according to Type I extreme value. If a consumer instead chooses the outside option, they obtain the indirect utility ε_{i0t}^d.[18]

Each consumer i in period t chooses the vehicle j or outside option that maximizes her payoff specified in equation (3). With our distribution assumption on ε_{ijt}^d, the conditional probability that consumer i from age group d chooses vehicle j in period t takes the logit form:

$$\widetilde{\mathfrak{s}}_{ijt}^d\left(\boldsymbol{\delta}_t^d, v_i, \sigma\right) = \frac{\exp\left(\delta_{jt}^d + \sigma v_i p_{jt}\right)}{1 + \sum_{j \in \mathcal{J}_t} \exp\left(\delta_{jt}^d + \sigma v_i p_{jt}\right)} \tag{4}$$

where

$$\delta_{jt}^d = \alpha^d p_{jt} + \beta^d A_{jt}^d + \boldsymbol{x}_{jt}' \boldsymbol{\gamma}^d + \vartheta_j^d + \varsigma_t^d + \nu_{gt} + \xi_{jt}^d \tag{5}$$

is the mean utility for product j for individuals in age group d in period t and $\boldsymbol{\delta}_t^d = \left\{\delta_{jt}^d\right\}_{j \in \mathcal{J}_t}$ is the vector of all vehicles' mean utilities for age group d in period t. Integrating out the idiosyncratic deviations from the mean utility gives the unconditional choice probability for vehicle j in period t from age group d, which is equivalent to its aggregate market share:

$$\mathfrak{s}_{jt}^d\left(\boldsymbol{\delta}_t^d, \sigma\right) = \int \widetilde{\mathfrak{s}}_{ijt}^d\left(\boldsymbol{\delta}_t^d, v_i, \sigma\right) \phi\left(v_i\right) dv_i \tag{6}$$

where $\phi\left(\cdot\right)$ is the pdf of the standard normal distribution. This demand model is flexible enough to capture the impact of pricing and advertising on demand, regardless of which view one takes on advertising.

[18] We normalize the mean utility of the outside option to zero in estimation, but in our counterfactual analysis we consider the impact of the (partial) advertising bans on the used car market and its resulting impact on the value of the outside option. We will discuss this in detail in Section 7.1.1.

4.2 Supply

Car manufacturers determine vehicle advertising and prices sequentially. Profit-maximizing retail prices depend on individuals' current advertising exposure states across different vehicles. Advertising decisions made today affect the pricing equilibrium in the subsequent period. Below, we describe the game in detail.

4.2.1 Pricing Decision

As defined in Section 3.3, individuals' advertising exposure states for vehicle j depend on past advertising exposure flows, approximately according to $A^d_{jt} \approx \delta A^d_{j,t-1} + a^d_{j,t-1}$.[19] Because individuals' advertising stocks depend only on past advertising, we assume firms take consumers' current exposure states as given when making pricing decisions, as in Dubois et al. (2018). Conditional on the vector of individuals' advertising exposure states $\boldsymbol{A}^d_t = \left\{A^d_{jt}\right\}_{j\in\mathcal{J}_t}$, firm f determines the retail prices of its vehicles that maximize its flow profit:

$$\Pi_{ft} = \sum_{j\in\mathcal{J}_{ft}} (\hat{p}_{jt} - c_{jt}) \left(\sum_{d\in\mathcal{D}} \mathfrak{s}^d_{jt}(\boldsymbol{p}_t, \boldsymbol{A}^d_t) M^d_t \right), \tag{7}$$

where $\mathcal{J}_{ft}$ denotes the set of vehicles offered by firm f and M^d_t denotes the market size (i.e., number of potential consumers of age group d).[20] Variables $\hat{p}_{jt}$ and c_{jt} denote, respectively, the retail price and marginal cost of producing vehicle j. The market share of vehicle j among individuals in age group d is $\mathfrak{s}^d_{jt}$ when the vehicle retail prices – net of the environmental subsidies (or penalties) – are given by $\boldsymbol{p}_t$ and the individuals' advertising exposure states are given by $\boldsymbol{A}^d_t$. According to the French feebate system, there is a financial rebate to consumers who buy less polluting vehicles, and an extra fee to consumers who buy more polluting vehicles. The exact amount of rebate or fee varies across vehicles and over time. We denote by τ_{jt} the rebate (or fee) applied to vehicle j. The net price of vehicle j for consumers p_{jt} is equal to the sum of the vehicle's retail price set by the car manufacturer $\hat{p}_{jt}$ and the feebate τ_{jt}: $p_{jt} = \hat{p}_{jt} + \tau_{jt}$. The value of τ_{jt} is positive in the case of a penalty (fee) and is negative in case of a subsidy (rebate).[21]

Car manufacturers have market power in this market and set their prices taking into account

[19] See equation (2).

[20] We set the market size M^d_t based on the assumption that adults on average purchase a vehicle every four years. Because we have quarterly sales data, this involves dividing the total population of an age group pop^d_t by $4 \times 4 = 16$.

[21] Depending on the vehicle j and period t, the value of τ_{jt} can be strictly positive, or strictly negative, or zero.

the demand and prices of rival manufacturers.[22] The price first-order conditions for firm f are:

$$\sum_{d\in D}\left(\mathfrak{s}_{jt}^{d}\left(\boldsymbol{p}_t,\boldsymbol{A}_t^d\right)+\sum_{j'\in\mathcal{J}_f}\left(p_{j't}-\tau_{j't}-c_{j't}\right)\frac{\partial\mathfrak{s}_{j't}^{d}\left(\boldsymbol{p}_t,\boldsymbol{A}_t^d\right)}{\partial p_{jt}}\right)M_t^d=0\quad\forall j\in\mathcal{J}_{ft}. \tag{8}$$

4.2.2 Advertising Decision

Advertising decisions proceed in two steps. First, the advertising slots are purchased. Then, the purchased slots are allocated across vehicles. Car manufacturers outsource the purchase of advertising slots to agencies, who negotiate with TV channels on their behalf. Data indicate that each brand contracts a single agency within a given year. The choice of agency and the brand-level advertising budget are determined at the firm level. Carmakers are multinational firms that select their advertising agencies centrally, with the French agency acting as the local office implementing these contracts. Local agencies therefore have no influence over the advertising budgets set by headquarters. Instead, they work with the local branches (brands) to secure the best advertising inventories within the given budgets and to determine the local marketing strategy, including which vehicle models to promote in the French market.

Formally, let $\widetilde{T}_{bst}$ be the duration in minutes of slot s that the agency buys for brand b in quarter t. The agency's problem is to choose the slots to buy for the brand to maximize the brand's total advertising exposure subject to the budget constraint:

$$\begin{aligned}&\max_{\widetilde{T}_{bst}\geq 0,\,\forall s\in\mathcal{S}_t}\sum_{d\in\mathcal{D}}\widetilde{a}_{bt}^{d}M_t^d=\sum_{d\in\mathcal{D}}\sum_{s\in\mathcal{S}_t}w_{st}^{d}\widetilde{T}_{bst}M_t^d\\&\text{subject to}\sum_{s\in\mathcal{S}_t}\kappa_{bt}\widetilde{T}_{bst}\leq e_{bt}\end{aligned} \tag{9}$$

where κ_{bt} denotes the advertising cost per minute for brand b. The term $\widetilde{a}_{bt}^{d}$ is the total expected advertising exposure of an individual in age group d in quarter t to brand b, M_t^d is the number of potential consumers in group d, and $\mathcal{S}_t$ is the set of advertising slots in quarter t.[23]

The agency then works with the local branch (brand) to determine which vehicle model(s) to promote in a given quarter. This amounts to the brand allocating each purchased slot $\widetilde{T}_{bst}>0$ to a vehicle $j\in\mathcal{J}_{bt}$, where $\mathcal{J}_{bt}$ is the set of products for sale by brand b in period t. In doing so, it is effectively choosing a flow of advertising exposure $a_{jt}^{d}(\boldsymbol{T}_{jt})$, where $\boldsymbol{T}_{jt}=\{T_{jst}\geq 0\}_{s\in\mathcal{S}_{bt}}$ and $\mathcal{S}_{bt}=\left\{s\in\mathcal{S}_t:\widetilde{T}_{bst}>0\right\}$, for each vehicle taking the viewing probabilities of each slot $\left\{w_{st}^{d}\right\}_{d\in\mathcal{D}}$ as given. Note that the agency chooses the set of advertising slots for the brand, $\widetilde{\boldsymbol{T}}_{bt}=\left\{\widetilde{T}_{bst}>0\right\}_{s\in\mathcal{S}_t}$,

[22]Note that individuals' advertising exposure states at period t do not depend on the future advertising exposure flows $t'>t$. When choosing vehicle prices for period t, firms observe all the past advertising exposure flows $t''<t$.

[23]Note that the advertising agency's audience-maximization program aligns with the advertiser's cost-minimization objectives (see Zhang, 2024). Both are simplifications of the complicated negotiations between agencies and TV channels.

while the brand allocates these slots across vehicles, i.e., determining $\{\boldsymbol{T}_{jt}\}_{j\in\mathcal{J}_{bt}}$, where $\boldsymbol{T}_{jt} = \{T_{jst} \geq 0\}_{s\in\mathcal{S}_{bt}}$.

We assume that a brand maximizes its expected return in the next period when making its advertising slot allocation decision.[24] The brand's objective function can be written as:

$$\max_{T_{jst}\geq 0, \forall j\in\mathcal{J}_{bt}, \forall s\in\mathcal{S}_{bt}} \mathbb{E}[R_{b,t+1}] \quad \text{subject to} \quad \sum_{j\in\mathcal{J}_{bt}} T_{jst} \leq \widetilde{T}_{bst},\ \forall s \in \mathcal{S}_{bt} \tag{10}$$

Let $\boldsymbol{\theta}$ denote the vector of demand parameters, and $\mathcal{F}_{\xi^d_{t+1}}$ the distribution of next period's demand shocks $\xi^d_{j,t+1}$, known to vehicle brands. The expected return of brand b in period $t+1$, conditional on information available at time t, is given by

$$\mathbb{E}[R_{b,t+1}] = \mathbb{E}_{(\boldsymbol{\theta},\ \boldsymbol{A}^d_t,\ \mathcal{F}_{\xi^d_{t+1}}, \forall d\in\mathcal{D})}\left[\sum_{j\in\mathcal{J}_{b,t+1}} (\hat{p}_{j,t+1} - c_{j,t+1}) \left(\sum_{d\in\mathcal{D}} \mathfrak{s}^d_{j,t+1}(\boldsymbol{p}_{t+1}, \boldsymbol{A}^d_{t+1}, \boldsymbol{X}_{t+1}) M^d_{t+1}\right)\right] \tag{11}$$

where $\boldsymbol{A}^d_{t+1} = \left\{A^d_{j,t+1}\right\}_{j\in\mathcal{J}_{t+1}}$ depends on the current stock of advertising $\boldsymbol{A}^d_t$, as well as the slot allocation decisions in current period $\boldsymbol{T}_t = \{\boldsymbol{T}_{jt}\}_{j\in\mathcal{J}_t}$, as specified in equations (1) and (2). The matrix $\boldsymbol{X}_{t+1}$ summarizes all the vehicle characteristics in period $t+1$. The demand shocks evolve according to the AR(1) process $\xi^d_{j,t+1} = \lambda_0 + \lambda_1 \xi^d_{j,t} + \tilde{\xi}^d_{j,t+1}$, where $\tilde{\xi}^d_{j,t+1}$ is an i.i.d. innovation. When allocating advertising slots, brands take into account how their current choices $\left\{\boldsymbol{T}^*_{jt}\right\}_{j\in\mathcal{J}_{bt}}$ influence the future equilibrium price vector $\hat{\boldsymbol{p}}^*_{t+1} = \left\{\hat{p}^*_{j,t+1}\right\}_{j\in\mathcal{J}_{t+1}}$.

4.3 Equilibrium

Individuals' current advertising exposure states $\boldsymbol{A}^d_t$ depend on their previous exposure states $\boldsymbol{A}^d_{t-1}$ and the advertising flows from the previous period $\boldsymbol{a}^*_{t-1} = \left\{a^*_{j,t-1}(\boldsymbol{T}^*_{j,t-1})\right\}_{j\in\mathcal{J}_{t-1}}$, as specified in equations (1) and (2). The equilibrium retail prices of vehicles, $\hat{\boldsymbol{p}}^*_t$, and the equilibrium advertising slot allocations, $\boldsymbol{T}^*_t = \left\{\boldsymbol{T}^*_{jt}\right\}_{j\in\mathcal{J}_t}$, are both functions of individuals' current advertising exposure states $\boldsymbol{A}^d_t$, as characterized by equations (8), (10) and (11). Individuals' vehicle demands, $\boldsymbol{\mathfrak{s}}^d_t$, depend on their advertising exposure states, $\boldsymbol{A}^d_t$, and the equilibrium vehicle prices, $\hat{\boldsymbol{p}}^*_t$. Formally, for each period t,

$$\begin{aligned}
&\boldsymbol{A}^d_t \equiv \boldsymbol{A}^d_t(\boldsymbol{A}^d_{t-1}, \boldsymbol{a}^*_{t-1}(\boldsymbol{T}^*_{t-1}), \boldsymbol{w}^d_{t-1}, \forall d\in\mathcal{D}) \\
&\hat{\boldsymbol{p}}^*_t \equiv \hat{\boldsymbol{p}}_t(\boldsymbol{X}_t, \boldsymbol{c}_t, \boldsymbol{\tau}_t, M^d_t, \boldsymbol{A}^d_t, \boldsymbol{\xi}^d_t, \forall d\in\mathcal{D}) \\
&\boldsymbol{T}^*_t \equiv \boldsymbol{T}_t(\boldsymbol{w}^d_t, \boldsymbol{\kappa}_t, \boldsymbol{e}_t, \boldsymbol{X}_{t+1}, \boldsymbol{c}_{t+1}, \boldsymbol{\tau}_{t+1}, M^d_{t+1}, \boldsymbol{A}^d_t, \mathcal{F}_{\xi^d_{t+1}}, \forall d\in\mathcal{D}) \\
&\boldsymbol{\mathfrak{s}}^d_t \equiv \boldsymbol{\mathfrak{s}}^d_t(\boldsymbol{A}^d_t, \hat{\boldsymbol{p}}^*_t, \boldsymbol{\tau}_t, \boldsymbol{X}_t, \forall d\in\mathcal{D}).
\end{aligned} \tag{12}$$

[24] We do not solve a dynamic advertising game, as vehicle attributes evolve in response to regulatory and technological shocks (Remmy, 2026; Barwick et al., 2024; Wollmann, 2018) that brands cannot anticipate over an infinite horizon.

The advertising viewing probabilities $\boldsymbol{w}_t^d$, slot prices $\boldsymbol{\kappa}_t$, and budgets $\boldsymbol{e}_t$ are treated as exogenous. Viewership and slot prices are unlikely to change as long as the structure of the media and agency markets remains unchanged, while advertising budgets are determined centrally by headquarters across countries and reflect competition among vehicle brands. These factors are therefore unlikely to be affected by advertising bans that are limited to the French market and apply uniformly across brands. One potential concern is that car manufacturers may respond to advertising bans by reducing their marketing budgets. However, we verify with our estimated model that no unilateral reduction in advertising expenditure is profitable. When determining the equilibrium advertising slot allocations $\boldsymbol{T}_t^* = \left\{\boldsymbol{T}_{jt}^*\right\}_{j\in\mathcal{J}_t}$, we assume that brands observe next period's vehicle characteristics and marginal costs of vehicle production. This assumption is justified by the fact that vehicle production typically precedes the marketing stage. In addition, brands are assumed to have full information about the next period's feebate policy and market size when choosing current-period advertising slot allocations.

We do not need to solve equation (9) (the agencies' problem) to simulate the effects of vehicle advertising bans. An outright ban eliminates the need for any advertising decisions, and partial bans are unlikely to alter the contracts between vehicle brands and their agencies.[25] Partial advertising bans restrict the set of vehicles eligible for advertising. As a result, brands must reoptimize the allocation of their purchased advertising slots over different feasible sets of vehicles induced by each ban. We describe in Section 7.1.2 how we solve the advertising slot allocation games, which feature multiple equilibria, using equations (10) and (11).

Following the literature (e.g., Reynaert, 2020, Durrmeyer, 2022), we assume the existence of a unique Bertrand–Nash equilibrium in vehicle prices. However, we allow for the possibility of multiple equilibria in the advertising slot allocation decisions.

5 Estimation and Identification

The vector of demand parameters to be estimated is given by:

$$\boldsymbol{\theta} = \left(\left\{\alpha^d, \beta^d, \boldsymbol{\gamma}^d, \left\{\vartheta_j^d\right\}_{j\in\mathcal{J}} \{\varsigma_t^d\}_{t\in\mathcal{T}}, \right\}_{d\in\mathcal{D}}, \{\nu_{gt}\}_{g\in\mathcal{G},t\in\mathcal{T}}, \sigma, \lambda_0, \lambda_1\right). \tag{13}$$

To estimate $\boldsymbol{\theta}$, we follow the approaches detailed in Conlon and Gortmaker (2020). Car manufacturers likely observe the vehicle-year-quarter-specific demand shocks ξ_{jt}^d prior to setting their prices p_{jt}. We rely on strong instruments, the Differentiation IVs recommended by Gandhi and Houde (2019) and applied in Barwick et al. (2024) and Almagro et al. (2024), to identify the price coefficients α^d. These instruments address the weak IV problem of using standard BLP instruments.

[25] The set of advertising slots $\widetilde{\boldsymbol{T}}_{bt} = \left\{\widetilde{T}_{bst}\right\}_{s\in\mathcal{S}_t}$ may change if car manufacturers adjust their advertising budgets for the French market under the partial bans. We consider this possibility in Section 7.1.2 (footnote 40).

Let $d^l_{jj',t}$ denote the distance in the value of product characteristic l between vehicles j and j' in in period t. This distance is defined as the absolute difference normalized by the standard deviation of that characteristic.[26] We then define a closeness weight, $z^l_{jj',t} = 1/(1 + d^l_{jj',t})$ which captures the proximity between vehicles j and j' in characteristic l. Let $\boldsymbol{z}_{j't}$ denote the vector of closeness weights across all characteristics l. Using these weights, we construct three sets of instruments: (i) sum of the closeness weights for all rival brands' products $\sum_{j' \in \mathcal{J}_t \setminus \mathcal{J}_{bt}} \boldsymbol{z}_{j't}$; (ii) sum of closeness weights for all rival brands' products in the same segment $\sum_{j' \in \mathcal{J}_{gt} \setminus \mathcal{J}_{bgt}} \boldsymbol{z}_{j't}$, where $\mathcal{J}_{bgt}$ is the set of products sold by brand b in segment g in period t; and (iii) sum of closeness weights for all other products of the same brand in the same segment $\sum_{j' \in \mathcal{J}_{bgt} \setminus \{j\}} \boldsymbol{z}_{j't}$.

These instruments, together with a vehicle's own characteristics $\boldsymbol{x}_{jt}$, constitute valid instruments for the price variable p_{jt} under the same timing assumption of the BLP IVs: the characteristics of a vehicle are chosen well before the realization of vehicle-quarter specific demand shock ξ^d_{jt}. On the other hand, characteristics of vehicles are observed before (and likely to influence) their pricing and advertising decisions. Table A1 in the Online Appendix shows regressions that correspond to the first stage of our instrumental variables approach. The estimates indicate that our Differentiation IVs explain the variation in vehicle prices well.

Identification of the advertising parameters β^d exploits rich variations in exposure to advertising at the vehicle-model level across consumer age groups and over time. As discussed in Section 3.5, implementing sophisticated, high-frequency targeted advertising in response to transient demand shocks is infeasible due to the institutional constraints of the ad-buying process in the French TV market. Moreover, data indicate that advertising for new vehicle models begins before any sales are observed, confirming that advertising campaigns are planned prior to the realization of age–vehicle–quarter–year-specific demand shocks ξ^d_{jt}. While targeting of advertising based on predictable demand shocks that are age- and/or period-specific is plausible, the reduced-form analysis in Section 3.5 shows that interacting consumer age groups with key vehicle characteristics, product fixed effects, and time fixed effects largely controls for such targeting.

Endogeneity of the advertising variables A^d_{jt} may still arise due to persistence in demand shocks: $\xi^d_{jt} = \lambda_0 + \lambda_1 \xi^d_{j,t-1} + \tilde{\xi}^d_{jt}$. Although vehicle brands do not observe ξ^d_{jt} when choosing $\boldsymbol{T}_{j,t-1}$, which determines A^d_{jt} (as specified in equations (1) and (2)), they do observe $\xi^d_{j,t-1}$ at the time of choosing $\boldsymbol{T}_{j,t-1}$. We isolate the econometric unobservable (which is unobserved by both brands and econometricians) by taking the pseudo-difference of the demand shocks: $\tilde{\xi}^d_{jt} = \xi^d_{jt} - \lambda_0 - \lambda_1 \xi^d_{j,t-1}$, similar to the approach taken in Grennan (2013), Sweeting (2013), and Lee (2013).

To estimate the parameter vector $\boldsymbol{\theta}$, we employ a two-step GMM procedure to obtain an efficient weighting matrix. The GMM objective function is constructed using $\tilde{\xi}^d_{jt}$ rather than ξ^d_{jt}. The AR(1) parameters λ_0 and λ_1 are identified through the time-series variation in ξ^d_{jt}, which comes from persistence in residual demand shocks after controlling for age-vehicle, age-year-quarter and

[26] For example, the distance in fuel cost between vehicles j and j' is calculated as the absolute difference in their fuel costs divided by the standard deviation of the fuel cost variable.

segment-year-quarter fixed effects, as well as observed vehicle attributes (weight, horsepower, fuel cost and body style) interacted with consumer age groups. To reduce the dimensionality of the nonlinear search over parameters, we first use the within transformation according to Conlon and Gortmaker (2020) to avoid directly estimating the 908 age-vehicle fixed effects ϑ_j^d. We then concentrate out the remaining linear parameters so that nonlinear search is only over the heterogeneity in price sensitivity σ and the AR(1) coefficients λ_0 and λ_1.[27] We compute robust standard errors using the standard asymptotic formula. We use the Gauss-Hermite Quadrature (GHQ) method to numerically approximate the integral in equation (6). Conlon and Gortmaker (2020) find that the GHQ method is more accurate and faster than to pseudo-Monte Carlo methods when the number of random coefficients is small.

We estimate the marginal cost parameters c_{jt} by inverting the system of price first-order conditions given by equation (8). Although we explore quarterly vehicle sales data by age group, we assume that the marginal cost of a vehicle is identical across age groups. We obtain an estimate of the marginal cost parameter by finding the value of c_{jt} that equates the population-weighted average of equation (8) to zero.

6 Model Estimates

6.1 Demand Estimates and Elasticities

We present the estimated coefficients of the model in equation (3) in Table 2, along with the average product-level price and advertising elasticities of demand by age group in Table A2 in the Online Appendix.[28] The vast majority of the parameter estimates are statistically significant. The estimated price and advertising parameters suggest that consumers become less price-sensitive and less advertising-responsive with age.[29] The average elasticities are weighted by market shares. The average own-price elasticities range from -2.21 to -5.37, while the average own-advertising elasticities range from 0.41 to 0.46, depending on the consumer age group. Given the large number of products in our sample (227 vehicles), Figure A16 in the Online Appendix plots the distribution of the estimated own-price and own-advertising elasticities. We note that the distribution of price elasticities is approximately normal. In contrast, the distribution of advertising elasticities is clearly right-skewed: the mean is pulled upward by a small number of high-elasticity products, while the median elasticity is substantially smaller than the mean. Our estimates of own-price elasticities are comparable to the values reported in D'Haultfœuille et al. (2019) and Grieco et al. (2024). The cross-price elasticities between vehicles are all positive.

[27] The linear parameters we concentrate out are the price coefficients α^d, advertising coefficients, β^d, effects of product characteristics $\boldsymbol{\gamma}^d$, age-year-quarter fixed effects ς_t^d, and segment-year-quarter trends ν_{gt}.

[28] The analytical formulas for the price and advertising elasticities of demand are shown in Section B.

[29] Although we find that consumers aged below 25 are less price-sensitive than those aged 25 to 34, this result should be interpreted with caution, as parents or grandparents may purchase vehicles on behalf of their children or grandchildren aged under 25.

Table 2: Demand Estimates

	All ages	Ages 0-24	Ages 25-34	Ages 35-49	Ages 50+
	(1)	(2)	(3)	(4)	(5)
Price: Mean		−2.555	−2.785	−2.217	−1.503
		(0.288)	(0.298)	(0.296)	(0.282)
Price: Std. Dev.	0.482				
	(0.217)				
Advertising exposure		0.037	0.025	0.016	0.010
		(0.002)	(0.001)	(0.001)	(0.000)
Weight		1.557	2.066	1.241	−0.074
		(0.424)	(0.443)	(0.433)	(0.413)
Power (kW)		2.237	2.586	1.995	1.173
		(0.337)	(0.346)	(0.347)	(0.329)
Fuel cost		−0.109	−0.121	−0.097	−0.063
		(0.015)	(0.016)	(0.015)	(0.015)
AR(1) coefficient for ξ^d_{jt}, λ_0	0.084				
	(0.013)				
AR(1) coefficient for ξ^d_{jt}, λ_1	0.650				
	(0.028)				
Age-body style fixed effects	Yes				
Age-vehicle fixed effects	Yes				
Age-year-quarter fixed effects	Yes				
Year-quarter-segment fixed effects	Yes				

Note: Robust standard errors in parentheses. Price is in €10,000. Weight is in 1,000 kilograms. Fuel costs are computed per 100 kilometers. Advertising exposure is measured according to equation (2), where T_{jst} is measured in minutes.

Our estimates of own-advertising elasticities are comparable to the figures reported in Barroso and Llobet (2012), who also estimate advertising demand elasticities for vehicles. Weighting plays an important role in shaping the average estimates, given the large number of heterogeneous products in this market. The unweighted average own-advertising elasticities are around 0.16 according to our demand estimates, which are similar to those reported in Honka et al. (2017) and Abi-Rafeh et al. (2026), but slightly higher than the elasticities documented in Shapiro (2018) and Shapiro et al. (2021). It is important to note that these prior studies (Honka et al., 2017; Shapiro, 2018; Shapiro et al., 2021; Abi-Rafeh et al., 2026) estimate the effects of *brand-level* advertising in markets such as banking, pharmaceuticals, and grocery items, which differ significantly from our estimation of *product-level* advertising effects in the vehicle industry. To confirm that the effects of advertising on demand are in the correct range, we first compute the marginal benefit of an additional minute of advertising a product on a brand's flow profit as predicted by our model (see the Online Appendix Section C for details). We then subtract from this the observed cost of an additional minute of advertising to obtain the net return to advertising. Figure A29 in the Online Appendix reports the distribution of estimated net returns on advertising. Consistent with brands' optimal advertising strategies, the distribution is centered around zero for advertised vehicle-quarters

and is predominantly negative for non-advertised ones. Cross-advertising elasticities are weak but negative.

In Tables A3 to A5 in the Online Appendix, we show the price and advertising cross-elasticities of demand split by CO_2 emissions, weight, and vehicle segment. The tables show the change in average demand for a vehicle in the category shown in the row with respect to a change in the average price (or advertising) of a vehicle in the category shown in the column. These tables provide additional evidence on the role of CO_2 emissions, weight, and segment in shaping consumer substitution patterns. If consumers view a particular characteristic as an important dimension of differentiation, substitution should be more limited across vehicles that differ along that dimension. Understanding these substitution patterns is important for evaluating the effectiveness of counterfactual advertising bans targeted at vehicles with specific attributes. One cannot solely consider a vehicle as non-polluting based on one of the three indicators. For example, the CO_2 emission level of an electric SUV that weighs above 3,000 kilograms is 0 g/km, although such a vehicle consumes significantly more energy than a lighter vehicle. On the other hand, EVs are generally heavier than their comparable vehicles powered by internal combustion engines.

Table A3 presents the average cross-elasticities by CO_2 emission levels. We split vehicles into three categories based on their emission level: low (less than 108 g/km), medium (between 108 g/km and 124 g/km), and high (more than 124 g/km). The estimates indicate important price competition involving low-emission vehicles, particularly EVs and hybrid vehicles emitting less than 108 g/km. When the price of a low-emission vehicle increases, consumers substitute not only toward other low-emission vehicles but, to an even greater extent, toward higher-emission vehicles. On the other hand, cross-advertising elasticities are remarkably similar across vehicles with different CO_2 emissions levels.

Table A4 presents the average cross-elasticities by vehicle weight. We split vehicles into three weight categories: less than 1,500 kilograms, 1,500–2,200 kilograms, and more than 2,200 kilograms. The estimates indicate that price increases for vehicles weighing between 1,500 and 2,200 kilograms generate larger substitution responses than comparable price changes for either lighter or heavier vehicles. Substitution involving vehicles weighing less than 1,500 kilograms is particularly weak. Cross-advertising elasticities exhibit little variation across vehicle weight levels.

Table A5 reports the average cross-elasticities by vehicle segment. The estimates indicate that substitution is strongest among mainstream vehicle types, particularly mini, small-family, and SUV models. Price increases for vehicles in the small-family segment generate the largest substitution responses across virtually all vehicle types, suggesting that these vehicles occupy a central position in consumer substitution patterns. By contrast, large-family vehicles, upper-class vehicles, and minivans exhibit substantially weaker substitution effects, both within and across segments. Cross-advertising elasticities vary little across vehicle types and reveal no clear systematic differences in advertising responsiveness.

Table 3: Average Price Elasticities of Demand

Age groups:	0-24	25-34	35-49	50+
Average own price elasticity	−4.521	−5.371	−4.023	−2.205
Average own price elasticity (zero ad stock)	−4.529	−5.398	−4.027	−2.183
Average cross price elasticity	0.006	0.014	0.012	0.004
Average cross price elasticity (zero ad stock)	0.004	0.010	0.009	0.003

Note: Elements in the table are calculated by taking the weighted average own- and cross-elasticities of vehicles within or across segments and brands, where each pair of vehicles is weighted by the product of their market shares. The term "zero ad-stock" refers to the situation where individuals' advertising exposure states are zero for all vehicles.

6.2 The Effects of Advertising on Price Elasticities

Table 3 shows the effect of advertising on the price elasticities of demand. We estimate the own- and cross-price elasticities of demand at the simulated market equilibrium with a zero advertising exposure stock for all vehicles. The results are compared to the price elasticities of demand at the baseline equilibrium with observed advertising. In the absence of advertising, consumers aged under 50 exhibit larger own-price elasticities, whereas consumers aged 50 and above exhibit smaller own-price elasticities. The differences relative to the baseline equilibrium are small across all age groups, indicating that advertising has only a limited effect on consumers' price sensitivity. Cross-price elasticities are larger in the equilibrium with advertising than in the no-advertising counterfactual for all age groups, suggesting that advertising increases substitution across vehicles. Nevertheless, the differences are modest in magnitude, indicating that the effect of advertising on substitution patterns is limited.

6.3 Markups and Margins of Car Manufacturers

Using the estimated marginal cost c_{jt}, we first calculate the profit margin for each vehicle, defined as $\hat{p}_{jt} - c_{jt}$. As expected, the most profitable segments are upper-class, large family, and SUVs. Next, we compute the product-level Lerner Index, which measures the markup of price over marginal cost: $L_{jt} = (\hat{p}_{jt} - c_{jt}) / \hat{p}_{jt}$. Our estimates suggest that average markups are highest for mini-vehicles, followed by small family cars and SUVs. This pattern is consistent with the more intensive advertising observed within these segments.

Table A6 in the Online Appendix reports the average markup by firm for the period 2012–2021, with values ranging from 0.16 to 0.37. On average, Renault exhibits the highest markup, followed by Suzuki, while TATA and Geely Group show the lowest.[30]

[30]Note that the firm Renault-Nissan-Mitsubishi was created in 2017 following the merger of Renault-Nissan and Mitsubishi. Therefore, in our sample, the firms Renault-Nissan and Mitsubishi existed until 2017, and were then replaced by Renault-Nissan-Mitsubishi. Similarly, in 2017, PSA acquired the brand Opel from GM. This acquisition did not result in a new firm, but rather caused a significant change in markup estimates of firms PSA and GM in 2017.

7 Counterfactual Analysis

We now explore several counterfactual scenarios involving vehicle advertising bans. Using our model and parameter estimates, we simulate vehicle pricing and demand under different advertising restrictions. We begin with an outright advertising ban, as advocated by some environmental activists.[31] In this scenario, individuals' exposure to vehicle advertising is entirely eliminated.

Next, we examine four partial advertising bans, each of which targets a specific subset of more-polluting vehicles. The first ban prohibits advertising for all SUVs. The second restricts advertising for high-emission vehicles. The third prohibits advertising for high-weight vehicles. The fourth targets non-electric and non-hybrid vehicles that exceed the same weight thresholds used in the high-weight ban. For the high-emission and high-weight criteria, we experiment with a range of alternative cut-off levels to define which vehicles fall into these categories.

Our analysis focuses on the short-term (one-year) effects of these advertising bans. Specifically, we assume that the bans are implemented in the last quarter of 2017, with their effects realized throughout 2018.[32] We then evaluate the environmental and welfare impacts of each of these possible bans.

7.1 The Environmental Impacts of Banning Vehicle Advertising

7.1.1 Outright Advertising Ban and Spillovers to the Used-Car Market

The outright advertising ban effectively sets individuals' advertising exposure flows to zero from the start of the ban (i.e., $a_{jt}^d = 0, \forall d, j$). In this counterfactual scenario, there is no need to solve for a new advertising equilibrium. Instead, we directly update the advertising exposure stocks under the ban using the ad stock formula in equation (2) with $a_{jt}^d = 0$. We then find the market equilibrium vehicle prices and sales using the demand and pricing model specified in Section 4, taking the updated advertising exposure stocks as given. A remaining challenge is to assess whether, and to what extent, the demand for used cars responds to the advertising ban. This allows us to determine where the displaced demand flows—whether consumers substitute toward (potentially more polluting) used vehicles or instead choose the no-car option—in order to evaluate the environmental implications of the ban. To explore whether the policy reshapes consumers' choices between second-hand vehicles and the other outside options, we estimate a multinomial logit model and use it to assess how advertising affects demand for used vehicles relative to the no-car alternative.

Let $\{\overline{0}\}$ denote the outside option of not owning a car or keeping one's existing vehicle, and let $\mathcal{J}_{\underline{0}t}$ denote the set of all second-hand vehicle alternatives. The full outside-option set is therefore given by $\mathcal{J}_{0t} = \{\overline{0}\} \cup \mathcal{J}_{\underline{0}t}$. We denote by $\delta_{\overline{0}t}^d$ the mean utility of not purchasing a car (neither new

[31] Source: `https://www.whichcar.com.au/news/environmentalists-push-to-ban-all-car-advertising`.

[32] Assessing the long-term impacts of advertising bans is beyond the scope of this study, as it would require accounting for changes in the supply of available vehicle models and their characteristics.

nor second-hand) in year-quarter t which, for identification, is normalized to zero. The indirect utility associated with this option is therefore given by $u^d_{i\underline{0}t} = \varepsilon^d_{i\underline{0}t}$. The mean utility of purchasing a second-hand model $\underline{k}$ in period t is specified as

$$\delta^d_{\underline{k}t} = \tilde{\beta}^d A^d_{\underline{k}t} + \vartheta_{\underline{k}} + \varsigma^d_t + \xi^d_{\underline{k}t} \tag{14}$$

where $A^d_{\underline{k}t}$ denotes the advertising stock for model $\underline{k}$ in quarter-year t for age group d, capturing the spillover effect of advertising for new vehicles on demand in the second-hand market. In the 2018 data, 22 percent of second-hand vehicles are sufficiently recent to have a positive stock of advertising $A^d_{\underline{k}t}$, after excluding rare vehicles, which together make up less than one percent of total second-hand sales. The term $\vartheta_{\underline{k}}$ is a vehicle-model fixed effect, while ς^d_t is an age-group-quarter-year fixed effect. Because we observe used-vehicle sales for only a single year, these product and age-group-year-quarter fixed effects absorb virtually all variation in time-invariant vehicle characteristics, such as weight, power, and fuel costs. Finally, $\xi^d_{\underline{k}t}$ denotes the second-hand-model-specific demand shock observed by consumers but not by the econometrician. Given the mean utility $\delta^d_{\underline{k}t}$, the indirect utility of second-hand model $\underline{k}$ is $u^d_{i\underline{k}t} = \delta^d_{\underline{k}t} + \varepsilon^d_{i\underline{k}t}$.

Assuming the idiosyncratic tastes $\varepsilon^d_{i\underline{0}t}$ and $\varepsilon^d_{i\underline{k}t}$ are Type I extreme value, the conditional probability that an individual in age group d, who has decided not to purchase a new car, chooses second-hand model $\underline{k}$ in period t takes the following logit form:[33]

$$\widetilde{\mathfrak{s}}^d_{\underline{k}t}\left(\delta^d_{\underline{0}}, \left\{\delta^d_{\underline{k}t}\right\}_{\underline{k}\in\mathcal{J}_{\underline{0}t}}\right) = \frac{\exp\left(\delta^d_{\underline{k}t}\right)}{\exp\left(\delta^d_{\underline{0}t}\right) + \sum_{\underline{k}\in\mathcal{J}_{\underline{0}t}} \exp\left(\delta^d_{\underline{k}t}\right)} \tag{15}$$

Following Berry (1994), the parameters in equation (14) can be obtained by estimating the linear regression $\ln(\widetilde{\mathfrak{s}}^d_{\underline{k}t}) - \ln(\widetilde{\mathfrak{s}}^d_{\underline{0}t}) = \delta^d_{\underline{k}t}$, where $\widetilde{\mathfrak{s}}^d_{\underline{0}t} = 1 - \sum_{\underline{k}\in\mathcal{J}_{\underline{0}t}} \widetilde{\mathfrak{s}}^d_{\underline{k}t}$. Results indicate a positive spillover effect of advertising for new-vehicle models on the second-hand market. More specifically, a one percent increase in the advertising-exposure stock of a new-vehicle model raises demand for the corresponding second-hand model by approximately 0.05 to 0.18 percent, depending on the consumer age group.[34] These spillover elasticities are roughly an order of magnitude smaller than

[33]Note that this approach is similar in spirit to estimating a two-step nested-logit demand model in which consumers first decide whether to purchase a new car, and then—conditional on not purchasing a new one—decide whether to acquire a second-hand vehicle and, if so, which model. However, because we do not observe second-hand vehicle prices and have only one year of second-hand sales data, we cannot credibly estimate all parameters in the ideal two-step nested-logit model. Given these data limitations, we instead adopt a simplified approach in which the two steps of consumer choice are estimated separately. The second step — which captures choices across second-hand models and other alternatives — is estimated using one year (four quarters) of data and without information on second-hand vehicle prices. We then use the estimated inclusive value in the second step as the renormalized value of the outside option in the first step and rescale the age-product fixed effects accordingly, thereby linking the two separately estimated models for the counterfactual analysis. While simplified, this approach provides indicative evidence on how the advertising ban may shift demand within the second-hand market and toward not purchasing a car.

[34]Elasticities are computed as sales-weighted averages over the 22 percent of second-hand vehicle models

the own-advertising elasticities of demand for new vehicles.[35] Using these parameter estimates to project demand across second-hand models and other outside alternatives under an advertising ban, we obtain the change in total CO_2 emissions and in the total vehicle weight from second-hand vehicles.

The total CO_2 emissions and vehicle weight levels can be calculated according to the following formulas:

$$\text{Total } \text{CO}_{2,t} = \left(\sum_d \frac{\sum_j \text{CO}_{2,j} \times \mathfrak{s}_{jt}^d}{\sum_j \mathfrak{s}_{jt}^d} \times M_{nt}^d + \sum_d \frac{\sum_{\underline{k}} \text{CO}_{2,\underline{k}} \times \widetilde{\mathfrak{s}}_{\underline{k}t}^d}{\sum_{\underline{k}} \widetilde{\mathfrak{s}}_{\underline{k}t}^d} \times M_{st}^d + \overline{\text{CO}}_{2,used} \times M_{\overline{0}_1 t} \right) \times \text{average annual number of kilometers driven per individual} \quad (16)$$

$$\text{Total weight}_t = \sum_d \frac{\sum_j \text{weight}_j \times \mathfrak{s}_{jt}^d}{\sum_j \mathfrak{s}_{jt}^d} \times M_{nt}^d + \sum_d \frac{\sum_{\underline{k}} \text{weight}_{\underline{k}} \times \widetilde{\mathfrak{s}}_{\underline{k}t}^d}{\sum_{\underline{k}} \widetilde{\mathfrak{s}}_{\underline{k}t}^d} \times M_{st}^d + \overline{\text{weight}}_{used} \times M_{\overline{0}_1 t} \quad (17)$$

where M_{nt}^d denotes the number of individuals in age group d who purchase new cars, M_{st}^d denotes the number of individuals in age group d who purchase second-hand vehicles, and $M_{\overline{0}_1 t}$ denotes the number of individuals who keep their existing vehicles. $\text{CO}_{2,j}$ and $\text{CO}_{2,\underline{k}}$ denote the per-kilometer CO_2 emissions of vehicle models j and $\underline{k}$, respectively. Likewise, weight_j and $\text{weight}_{\underline{k}}$ denote the weights of vehicle models j and $\underline{k}$. Finally, $\overline{\text{CO}}_{2,used}$ and $\overline{\text{weight}}_{used}$ denote the sales-weighted average per-kilometer CO_2 emissions and vehicle weight, respectively, of used vehicles.[36] We simulate the values of $\mathfrak{s}_{jt}^d$, $\widetilde{\mathfrak{s}}_{\underline{k}t}^d$, M_{nt}^d, M_{st}^d under both the baseline and counterfactual scenarios. Under the assumption that the number of current car owners who choose to keep their existing vehicle remains unchanged between the baseline and counterfactual scenarios, $M_{\overline{0}_1 t}$ is held constant across scenarios and is equal to the total vehicle fleet minus M_{nt}^d and M_{st}^d in the baseline scenario.[37] Individuals in France drove an average of 12,508 kilometers in 2018 according to the French Statistical Data and Studies Department.[38] We use this number to obtain the total annual CO_2 emissions per vehicle.

The impact of an outright advertising ban on total CO_2 emissions and vehicle weight levels is reported in the first row of Table 4. The percentage changes are calculated as the proportional difference between the counterfactual and baseline totals, that is, (counterfactual total − baseline total)/(baseline total). The counterfactual total is derived from the model-simulated pricing equilibrium with zero advertising flows for all individuals from the beginning of the ban, while

with positive advertising-exposure stocks $A_{\underline{k}t}^d$.

[35] Because we omit transaction prices for second-hand purchases, this model likely underestimates the true sensitivity of second-hand demand to advertising. Nevertheless, Figure A17 in the Online Appendix compares the second-hand vehicle demands predicted by equation (15) with the observed sales in the data. The model provides a very good fit.

[36] For this we take the sales-weighted average per-kilometer CO_2 (in g/km) and weight of used cars in the dataset.

[37] An alternative assumption could be that the ratio of non-car owners to car owners choosing to keep their old car remains the same between the baseline and counterfactual scenarios. Both assumptions yield very similar quantitative results and lead to the same policy conclusions.

[38] Source: `https://www.statistiques.developpement-durable.gouv.fr`.

Table 4: Environmental Effects of Banning Vehicle Advertising

Percentage change in:	Total CO_2 emissions	Total vehicle weight
Outright ban	[-3.29, -3.29]	[-3.27,-3.27]
SUV ad ban	[-0.17, 0.38]	[-0.27,0.24]
High CO_2 ad ban	[-2.37, -2.32]	[-2.31,-2.26]
High weight ad ban	[-2.63, -2.61]	[-2.87,-2.86]
Non-electric/hybrid high weight ad ban	[-2.50, -2.46]	[-2.61,-2.57]

Note: All results [lower bound, upper bound] are percentage changes relative to the baseline equilibrium with no advertising ban. The high-CO_2 ad ban targets vehicles emitting more than 108g of CO_2 per km. The high-weight ad ban targets vehicles weighing more than 1,500 kg. The same weight threshold (1,500 kg) is applied in the non-electric/hybrid high-weight ad ban.

the baseline total is derived from our model-simulated advertising and pricing equilibrium in the absence of an advertising ban.[39] Advertising contributes to the expansion of the new-car market and also increases demand for second-hand vehicles by approximately one percent. A complete ban would reduce demand for both new and used vehicles, with more individuals choosing the no-car alternative instead. Overall, an outright ban is predicted to reduce total CO_2 emissions and total vehicle-weight levels by more than three percent.

7.1.2 Targeted Advertising Bans and Ad Slot Allocation Across Products

We next examine four partial advertising bans, each targeting a specific subset of polluting vehicles. These bans are unlikely to affect the overall set of advertising slots purchased by agencies but will alter how vehicle brands allocate those slots across different models.[40] Below, we detail the algorithm used to establish approximate bounds on the equilibrium effects of partial advertising bans. We incorporate machine-learning methods into our equilibrium-search procedure, which substantially accelerates the computation of equilibrium pricing and advertising decisions.

[39]The observed stocks of advertising serve as the initial guess for determining the baseline advertising equilibrium using the algorithm described in Section 7.1.2 below. Figure A18 compares the model-predicted baseline sales of new cars with the observed sales in the data. The model fits the data very well. Figure A19 shows a histogram of the share of products predicted to be advertised that match the data by brand-year-quarter, excluding the trivial cases where no or all products are advertised. Again, the model predicts the correct products to be advertised in the majority of cases.

[40]It is highly unlikely that introducing a ban on advertising for a subset of polluting vehicles would alter the structure of the advertising market. TV broadcasters and advertising agencies serve advertisers from a wide range of industries, and partial vehicle advertising bans would not remove automobile manufacturers from the market altogether. Therefore, the set of advertising slots available to vehicle brands is not likely to change. Such changes would arise only if vehicle manufacturers adjusted their overall advertising budgets. However, advertising budgets depend primarily on competitive conditions in the vehicle market, which should not change in response to advertising bans that apply uniformly across brands. Moreover, we focus on the short-run (one-year) impact of advertising bans, taking as given the set of vehicle models offered by each brand. Nevertheless, as a robustness check, we re-simulate the equilibrium effects of the partial advertising bans under scenarios in which vehicle brands' advertising budgets increase or decrease by 10 percent. The results, reported in Figure A23 in the Online Appendix, indicate that the qualitative conclusions and policy implications of our analysis are robust to these changes.

An equilibrium is defined as in Section 4.3. Any change in the allocation of advertising slots leads to new vehicle prices and market shares in the subsequent period.[41] In 2018, 27 of the 32 brands advertised at least one product, with half of these brands selling more than 8 products. Our counterfactual scenarios have many possible advertising allocations per time period, each being a potential market equilibrium. For example, the SUV advertising ban has over 2.95×10^{23} possible advertising allocations per time period. Recognizing the possibility of multiple equilibria, we form approximate bounds on the outcomes across different counterfactual scenarios.

In order to obtain an approximate bound on the effect of a partial advertising ban on total CO_2 emissions, we initialize the search for an equilibrium allocation of advertising slots from two extreme initial allocations: (i) all brands allocate slots to their vehicles emitting the least CO_2 per kilometer, and (ii) all brands allocate slots to their vehicles emitting the most CO_2 per kilometer but not banned for advertising. The intuition behind this approach is that these extreme initial allocations will produce equilibria that approximately bound the effect of the ban on CO_2 emissions.

To obtain the impact of the ban on total weight, we repeat the above approach but instead use initial allocations based on product weights. As we will discuss in Section 7.2, we also aim to quantify the effects of the bans on consumer welfare and profits. For consumer welfare, we assume brands allocate advertising to their unbanned products with the highest and lowest mean utility in the absence of advertising.[42] Finally, for total firm profits, we use extreme initial allocations based on profit margins.[43] In total, this results in eight extreme initial allocations, which together form approximate bounds on total CO_2 emissions, weight, consumer welfare and profits.[44]

For each initial allocation and under all counterfactual scenarios, we first solve for the new price equilibrium and market shares under these advertising levels. We then proceed to search for the equilibrium advertising allocations in which no brand has an incentive to deviate from its choice. Brands maximize their expected returns in the next period $\mathbb{E}[R_{b,t+1}]$, defined in equation (11), when deciding on the allocation of advertising slots $\{\boldsymbol{T}_{jt}\}_{j\in\mathcal{J}_{bt}}$. They deviate from the current allocation if an alternative allocation yields higher expected returns. We find the equilibrium advertising slot allocations using an algorithm proposed by Lee and Pakes (2009) and Wollmann (2018). In short, the program assumes an ordering of decisions based on brands' market shares. The first brand chooses

[41] We define a period as a quarter.

[42] We define the mean utility of a new product j for an individual in age group d, absent advertising, as

$$\widehat{\delta}^d_{jt} = \alpha^d p_{jt} + \boldsymbol{x}'_{jt}\boldsymbol{\gamma}^d + \vartheta^d_j + \varsigma^d_t + \nu_{gt} + \xi^d_{jt} \tag{18}$$

For this set of initial allocations, for each brand, we initialize advertising allocations to the vehicle(s) with the highest and lowest values of $\sum_{d\in\mathcal{D}}\widehat{\delta}^d_{jt}M^d_{nt}/\sum_{d\in\mathcal{D}}M^d_{nt}$.

[43] These extreme allocations for firms' profits are motivated by our finding that advertising primarily affects market shares rather than vehicle pricing.

[44] As a robustness check, we additionally tested 50 random allocations of advertising slots as starting points in the equilibrium search, compared to the extreme allocations. The resulting equilibrium outcomes from these 50 random starting allocations, as reported in Figures A24 and A25 in the Online Appendix, closely align with the equilibrium bounds derived from the extreme starting allocations. This confirms that the equilibrium bounds established through our search algorithm are not sensitive to the starting points.

the allocation of its advertising slots that best responds to the allocations of all other brands.[45] The second brand similarly chooses its best response, taking as given the first brand's best response. The third brand also responds in this manner, taking the first and second brands' allocations as given. This cycle continues across all brands, with the program updating each brand's advertising allocations across vehicles until a full cycle is complete and no brand wishes to deviate further. The result is a simultaneous-move Nash equilibrium, $\{\boldsymbol{T}_t^*, \boldsymbol{A}_{t+1}^d(\boldsymbol{T}_t^*), \hat{\boldsymbol{p}}_{t+1}^*(\boldsymbol{T}_t^*), \mathfrak{s}_{t+1}^d(\boldsymbol{T}_t^*, \hat{\boldsymbol{p}}_{t+1}^*(\boldsymbol{T}_t^*))\}$.

Because computing the expected returns $\mathbb{E}[R_{b,t+1}]$ for each feasible advertising-slot allocation $\boldsymbol{T}_t = \{\boldsymbol{T}_{jt}\}_{j\in\mathcal{J}_t}$ requires taking a large number of random draws of $\tilde{\boldsymbol{\xi}}_{t+1}^d$, this procedure is extremely time-consuming.[46] To reduce the computational burden, we rely on a machine learning model to predict $\mathbb{E}[R_{b,t+1}]$ directly from a given allocation $\boldsymbol{T}_t$. Thus, in the equilibrium-search algorithm described above, when cycling across brands and evaluating the profitability of potential deviations in advertising choices, we use the machine-learning-predicted product-level returns rather than solving for the full price equilibrium for each candidate deviation. To train this machine learning model, we generate 100 samples, each of which is a copy of our data from 2018 but with a randomly assigned advertising allocation $\boldsymbol{T}_t$ for each product. For each sample, we compute the expected product-level returns by averaging over 200 independent draws of $\tilde{\boldsymbol{\xi}}_{t+1}^d$. To do this, for a given sample-draw combination, we solve for the price equilibrium and compute the associated product-level returns. Averaging over draws gives the expected returns from that advertising allocation. Using these generated expected returns, we train a random forest model. The variables we use in this model are the advertising stock of the car model, the product's mean utility for consumers in the absence of advertising, and the estimated marginal cost plus the observed feebate. Additionally, we use age and brand fixed effects. We use 100 trees with a maximum of 15 splits per tree. The model is trained on 80 of the samples and validated on the remaining 20. Figure A20 in the Online Appendix reports the out-of-sample prediction accuracy of this approach for these 20 samples. The machine learning model is able to accurately predict expected brand returns from a given advertising allocation, and can do so much faster than the full solution approach.

Note that vehicle brands' expected returns depend only on new-vehicle sales. They do, however, take into account whether and how their allocations of advertising slots affect overall demand for the outside option through equation (4). Starting from the extreme advertising-allocation starting values, we find the market-equilibrium allocations of advertising slots that yield the highest and lowest total CO_2 emissions and vehicle-weight levels for new vehicles sold using the machine learning model.[47] We then use model (15) and its parameter estimates to predict demand for

[45] Because we observe that brands advertise only a small number of vehicles per period, we further assume that brands continue to advertise the same total number of models in each counterfactual. This reduces the number of possible alternatives we need to consider when finding a brand's best response.

[46] Advertising decisions are made before $\xi_{j,t+1}^d$ is known. Brands therefore form expectations using the AR(1) process $\xi_{j,t+1}^d = \lambda_0 + \lambda_1 \xi_{j,t}^d + \tilde{\xi}_{j,t+1}^d$ and the distribution of the innovation $\tilde{\xi}_{j,t+1}^d$, which is assumed to be normally distributed. Its variance is estimated from the fitted AR(1) process.

[47] We confirm that the equilibrium outcomes are not sensitive to the search order (based on brands' market shares) or to the criteria used to rank vehicle pollution levels (whether by CO_2 emissions or by weight).

different second-hand models and for the no-car alternative under these two equilibrium advertising allocations (i.e., the allocations that produce the approximate upper and lower bounds for total CO_2 emissions and vehicle-weight levels of new vehicles sold). Finally, we obtain the approximate upper and lower bounds on *total* CO_2 emissions and vehicle-weight levels in market equilibrium under each partial advertising ban by aggregating emissions from new-vehicle purchases, second-hand purchases, and the residual group of individuals who retain their existing vehicles, following the aggregation formulas in equations (16) and (17).

We experiment with a broad set of alternative cut-off levels to determine which high-emission and high-weight vehicles should be banned from advertising. Figure A21 and Figure A22 in the Online Appendix summarize the resulting environmental impacts across 10 CO_2 thresholds and 12 weight thresholds for advertising bans. The whiskers show the range of potential environmental outcomes for each advertising ban, representing the smallest and largest total CO_2 emission and vehicle-weight levels in market equilibrium. These results indicate that relatively stringent thresholds—below 116 grams of CO_2 per kilometer for the emissions criterion or below 1,800 kilograms for the weight criterion—are necessary to generate meaningful environmental improvements. Setting the high-emission threshold to 108 grams of CO_2 per kilometer or the high-weight threshold to 1,500 kilograms yields environmental effects comparable to those of an outright advertising ban.

Table 4 compares the approximate bounds of the environmental outcomes for all four partial advertising bans—those banning advertising for SUVs, high-emission vehicles, high-weight vehicles, and non-electric/hybrid high-weight vehicles—to the effect of an outright advertising ban. We set the thresholds to 108 grams of CO_2 per kilometer for the high-emission vehicles and 1,500 kilograms for the high-weight vehicles. The same weight threshold (1,500 kilograms) is used for the non-electric/hybrid advertising ban. It is important to note that we propose a fast and relatively simple method for establishing approximate bounds on the environmental effects of vehicle advertising bans. Our algorithm simulates the equilibrium upper and lower bounds on total CO_2 emissions and total weight of vehicles sold under these bans. Other equilibria in which the total emissions and weight of vehicles sold fall outside these bounds may exist, but in Figures A24 and A25 in the Online Appendix, we show that the environmental effects from all equilibria from 50 random initial allocations fall within our approximate bounds. As shown in Table 4, most bounds are quite narrow, enabling clear policy conclusions.

7.1.3 Environmental Impacts: Outright versus Targeted Bans

The environmental impacts of advertising bans are generally positive, in the sense that they reduce total CO_2 emissions and the aggregate vehicle-weight level. The magnitude of these improvements, however, varies across policy designs. An outright advertising ban delivers the largest reductions in both total CO_2 emissions and aggregate vehicle weight. By contrast, banning advertising for SUVs yields only limited environmental benefits. In comparison, the high-CO_2 advertising ban (targeting vehicles emitting more than 108 grams of CO_2 per kilometer), the high-weight advertising ban

(targeting vehicles weighing more than 1,500 kilograms), and the non-electric/hybrid high-weight advertising ban effectively restrict advertising to electric and hybrid vehicles and light vehicles. These targeted bans generate reductions in total CO_2 emissions and aggregate vehicle-weight levels that are comparable in magnitude to those achieved by an outright advertising ban. The positive environmental effects of advertising bans are driven primarily by consumers switching from vehicle ownership (both new and used cars) to the no-car option. Advertising appears to persuade some consumers—who may not strictly require a vehicle—to purchase one.[48] This conclusion is robust to the alternative assumption regarding the share of current car owners who choose to retain their existing vehicle in the counterfactual scenarios (see footnote 37). Table A7 reports the corresponding environmental impacts of advertising bans. The results are broadly similar to those reported in Table 4.[49]

7.2 The Welfare Effects of Banning Vehicle Advertising

In Table 5, we compare the welfare consequences of the five advertising bans. We report the resulting changes in consumer surplus, firms' profits, feebate revenues, social benefits, and total welfare, measured in billions of euros. Our demand model is sufficiently flexible to capture the effects of advertising on individuals' vehicle choices, regardless of which view one takes on advertising. However, evaluating the welfare consequences of banning vehicle advertising requires taking a stance on the underlying role of advertising. We assume that vehicle advertising influences consumer choices without affecting their experienced utility, consistent with the persuasive view of advertising in Dixit and Norman (1978) and Dubois et al. (2018). This interpretation is supported by our finding that advertising induces some consumers—who may not strictly need a car—to purchase one. Under this view, consumers choose the product with the highest value of u^d_{ijt} or $\delta^d_{\underline{k}t}$ according to equations (3) and (14) (which include advertising), but their experience utility is based on their valuation of the product in the absence of advertising:

$$\widehat{u}^d_{ijt} = \alpha^d_i p_{jt} + \boldsymbol{x}'_{jt}\boldsymbol{\gamma}^d + \vartheta^d_j + \varsigma^d_t + \nu_{gt} + \xi^d_{jt} + \varepsilon^d_{ijt} \tag{19}$$

$$\widehat{u}^d_{i\underline{k}t} = \vartheta_{\underline{k}} + \varsigma^d_t + \xi^d_{\underline{k}t} + \varepsilon^d_{i\underline{k}t} \tag{20}$$

In more detail, the consumer's expected experience utility is given by:

$$\widehat{V}^d_{it}\left(\boldsymbol{A}_t, \boldsymbol{p}_t\right) = \mathbb{E}_\varepsilon\left[\max_{j\in\mathcal{J}_t\cup\mathcal{J}_{0t}} u^d_{ijt}\right] - \sum_{j\in\mathcal{J}_t} \widetilde{\mathfrak{s}}^d_{ijt}\left(u^d_{ijt} - \widehat{u}^d_{ijt}\right) - \sum_{\underline{k}\in\mathcal{J}_{\underline{0}t}} \widetilde{\mathfrak{s}}^d_{i\underline{k}t}\left(u^d_{i\underline{k}t} - \widehat{u}^d_{i\underline{k}t}\right) \tag{21}$$

[48] Advertising also influences consumer preferences and may distort them, as evidenced by the increase in consumers' experience utility under advertising bans (see Section 7.2).

[49] Table A8 additionally reports the corresponding welfare impacts of advertising bans. The results are broadly similar to those reported in Table 5.

Table 5: The Welfare Effects of Banning Vehicle Advertising

	Change in consumer welfare	Change in firm profits	Change in feebate revenue	Change in social benefit	Change in total welfare
Outright ban	[2.11, 2.11]	[-1.62, -1.62]	[0.03, 0.03]	[0.05, 0.05]	[0.57, 0.57]
SUV ad ban	[0.21, 0.60]	[-0.28, -0.04]	[0.02, 0.04]	[-0.01, 0.00]	[0.16, 0.42]
High CO_2 ad ban	[1.36, 1.40]	[-1.01, -0.96]	[-0.02, -0.01]	[0.03, 0.03]	[0.40, 0.42]
High weight ad ban	[1.69, 1.70]	[-1.20, -1.18]	[0.03, 0.03]	[0.04, 0.04]	[0.56, 0.58]
Non-electric/hybrid high weight ad ban	[1.64, 1.69]	[-1.08, -1.04]	[-0.02, -0.02]	[0.03, 0.04]	[0.60, 0.63]

Note: All results [lower bound, upper bound] are in billions of euros.

where $\mathbb{E}_\varepsilon\left[\max_{j\in\mathcal{J}_t\cup\mathcal{J}_{0t}} u^d_{ijt}\right] = \ln\left[\exp\left(I^d_{i0t}\right) + \sum_{j\in\mathcal{J}_t}\exp(\delta^d_{jt}+\sigma v_i p_{jt}+\psi^d_t)\right]$. The inclusive value for a second-hand buyer is given by $I^d_{i0t} = \ln\left[1+\sum_{\underline{k}\in\mathcal{J}_{0t}}\exp(\delta^d_{\underline{k}t})\right]$. The term ψ^d_t renormalizes the value of the outside option. It is equal to I^d_{i0t} in the baseline case, and remains unchanged in the counterfactual. Let $\boldsymbol{A}^c_t$ and $\boldsymbol{p}^c_t$ denote the equilibrium advertising exposure state and price vectors under the advertising bans. Denote by $\boldsymbol{A}^0_t$ and $\boldsymbol{p}^0_t$ the same under the baseline equilibrium without a ban. The change in consumer surplus is measured as $\int \frac{1}{\alpha^d_i}\left[\widehat{V}^d_{it}\left(\boldsymbol{A}^0_t,\boldsymbol{p}^0_t\right) - \widehat{V}^d_{it}\left(\boldsymbol{A}^c_t,\boldsymbol{p}^c_t\right)\right]\phi\left(v_i\right)dv_i$.

Starting from the eight initial allocations described in Section 7.1, we use the equilibrium search algorithm to compute the upper and lower equilibrium bounds for changes in the expected experienced utility of *new-car buyers* under each advertising ban. We then use model (15) and its parameter estimates to simulate the corresponding changes in the expected experienced utility of *second-hand buyers*, allowing us to obtain upper and lower equilibrium bounds for the *total* change in consumer surplus. The results are reported in the first column of Table 5. Additionally, we compute the change in firm profits and feebate revenue, which we report in columns 2 and 3.

Finally, we evaluate the change in social benefits as the reduction in the social cost of CO_2 emissions and vehicle weight. A recent report by the European Commission estimates the climate-change cost in the short- to medium-run (up to 2030) at €100 per metric ton of CO_2-equivalent (in 2016 euros).[50] Using these benchmark estimates, we quantify the reduction in the social cost of emissions under advertising bans by multiplying the resulting changes in total CO_2 emissions at the equilibrium bounds by €100 (in 2016 euros) per metric ton. In addition, a Dutch study estimates the external cost of vehicle weight at between €0.001 and €0.002 per additional kilogram per year.[51] Multiplying this estimate by the simulated change in aggregate vehicle weight under each advertising-ban scenario yields an estimate of the corresponding reduction in weight-related external costs. Adding this amount to the reduction in the social cost of emissions yields the total reduction in social costs. Results are reported in the fourth column of Table 5.

Upper and lower bounds of the changes in consumer surplus, firms' profits, feebate revenues, and social benefits correspond to eight different equilibrium bounds. We calculate the changes in

[50] See Van Essen et al. (2019).

[51] See Schroten et al. (2022).

total welfare in each of the eight market equilibria bounds and report the minimum and maximum changes in total welfare in the last column of Table 5.[52]

All five bans considered increase consumer welfare. Restricting advertising reduces firms' ability to distort consumer choices, bringing purchase decisions closer to consumers' optimal choices in the absence of advertising. On the other hand, all five bans reduce firms' aggregate profits because they decrease total new-vehicle sales, with the magnitude of the effect varying across policies. The effects of advertising bans on feebate revenues reflect the substitution patterns they induce across vehicles. The outright ban and the high-weight ban raise feebate revenues by reducing demand for highly subsidized EVs. In contrast, the high-CO_2 ban and the ban on non-electric/non-hybrid high-weight vehicles lower feebate revenues because they primarily reduce demand for highly polluting vehicles that incur environmental penalties under the feebate system. With the exception of the SUV ban, the other four bans generate modest but positive reductions in the social costs of vehicles. Overall, all five advertising bans considered produce net welfare gains.

Among the policies considered, an outright advertising ban generates the largest reductions in total CO_2 emissions and vehicle weight but also leads to substantial profit losses for firms, potentially limiting its practical feasibility. In contrast, a ban on advertising non-electric and non-hybrid vehicles weighing more than 1,500 kilograms achieves similar environmental benefits while generating the highest welfare gains, estimated at approximately €600 million.

8 Conclusion

This paper examines the potential environmental and welfare consequences of vehicle advertising bans within a structural IO framework. We develop a model of vehicle demand, pricing, and advertising that allows for heterogeneity in advertising responsiveness, price sensitivity, and preferences for vehicle attributes across consumer age groups. The model is estimated using detailed product-level and age-group-level data on vehicle sales and advertising in France from 2012 to 2021. Using the estimated model, we simulate equilibrium advertising allocations, vehicle prices, and demand under five counterfactual advertising policies: (i) an outright advertising ban, (ii) an SUV advertising ban, (iii) a high-CO2 advertising ban, (iv) a high-weight advertising ban, and (v) a ban on advertising non-electric and non-hybrid high-weight vehicles. Across the policies considered, the ban on advertising non-electric and non-hybrid vehicles weighing more than 1,500 kilograms emerges as the most effective targeted intervention. In addition to generating aggregate welfare gains that exceed those of a complete advertising ban, the policy achieves these gains through a different redistribution of surplus between consumers and firms, and delivers broadly comparable environmental improvements.

As the transition toward more fuel-efficient vehicles becomes increasingly urgent, environmental

[52]Our measure of total welfare does not include advertising spending, as it constitutes a transfer from car manufacturers to media content providers.

activists and policymakers worldwide are exploring policies that target vehicle advertising. France has already adopted legislation that will prohibit advertising for certain polluting vehicles from 2028 onwards. Our findings therefore provide timely evidence and clear policy guidance on the likely environmental and welfare effects of such interventions.

We view vehicle advertising bans as complementary to taxation rather than as a substitute for it. Nevertheless, comparing the effectiveness of these two policy instruments and examining how they interact represent promising direction for future research.

We develop a framework and an efficient algorithm for simulating the counterfactual effects of partial advertising bans that can be readily generalized and applied in other settings. One limitation of our analysis is its focus on the short-run effects of advertising restrictions. Evaluating the long-run consequences of such policies, including how vehicle manufacturers adjust their advertising budgets, product offerings, and vehicle attributes in response to advertising bans and other environmental regulations, remains an important question.

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Online Appendix to:

Environmental Impacts of Banning Vehicle Advertising

A Additional Figures and Tables

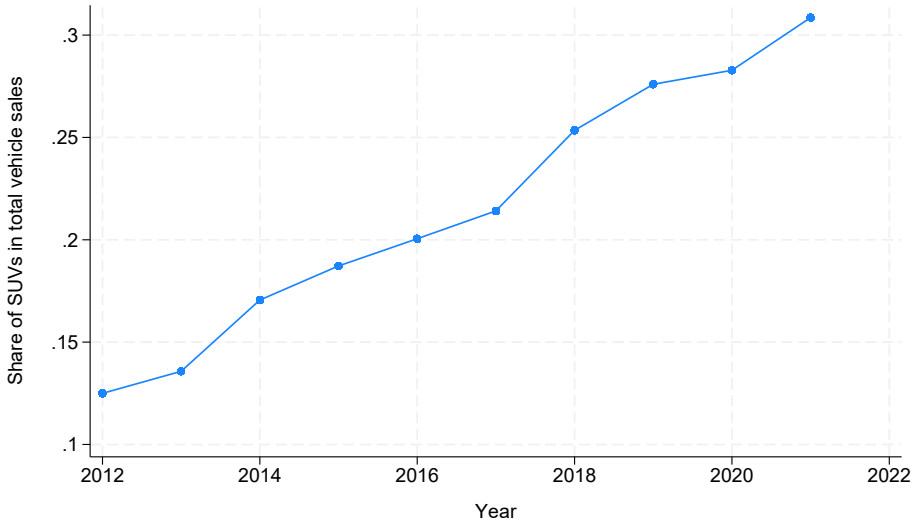


Figure A1: Share of SUVs in Total Vehicle Sales

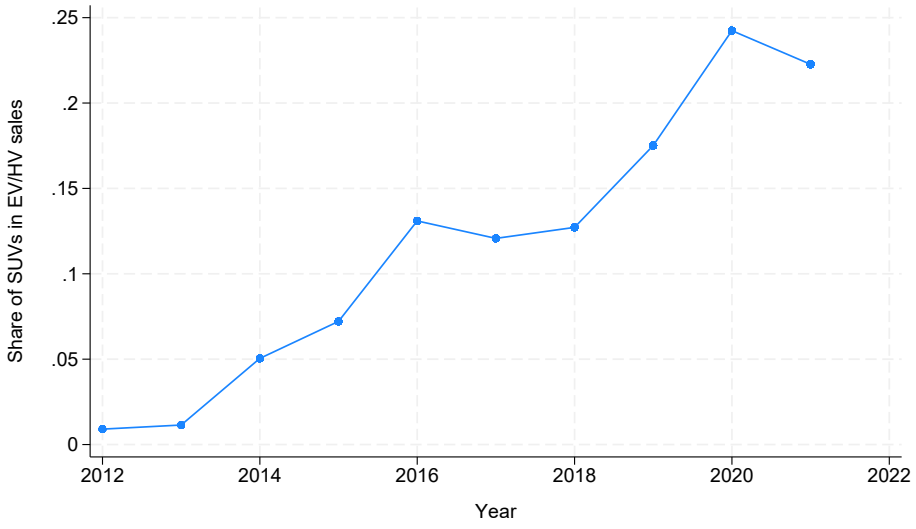


Figure A2: Share of SUVs in Electric and Hybrid Vehicle Sales

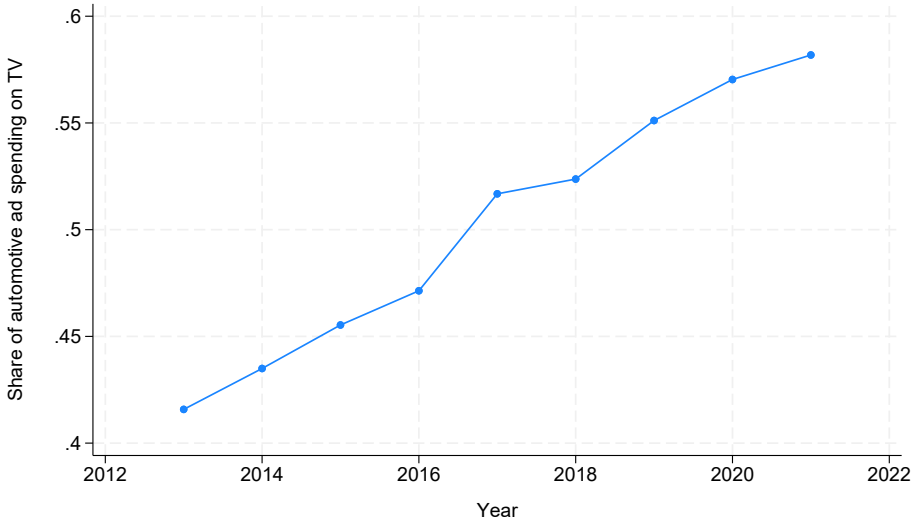


Figure A3: Share of Automotive Advertising Expenditure on TV (Total Spending on TV/Total Spending on All Media)

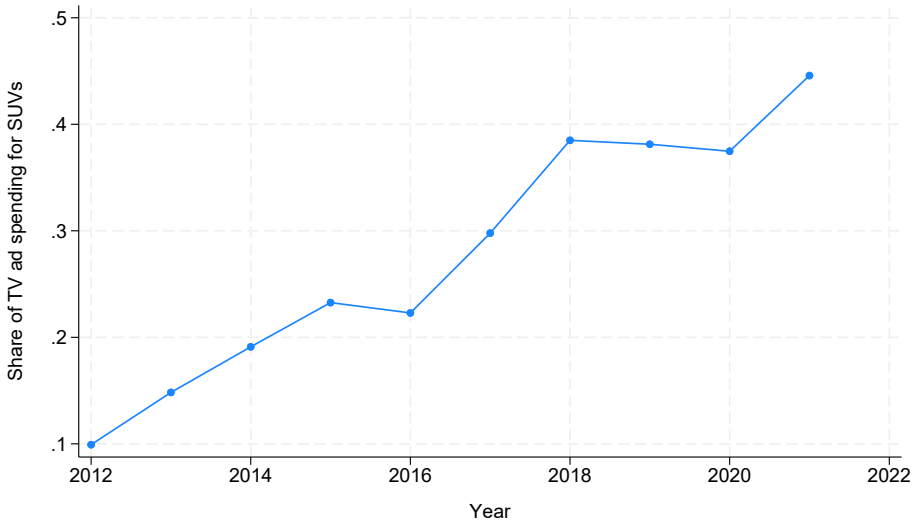


Figure A4: Expenditure Share of SUVs in TV Advertising

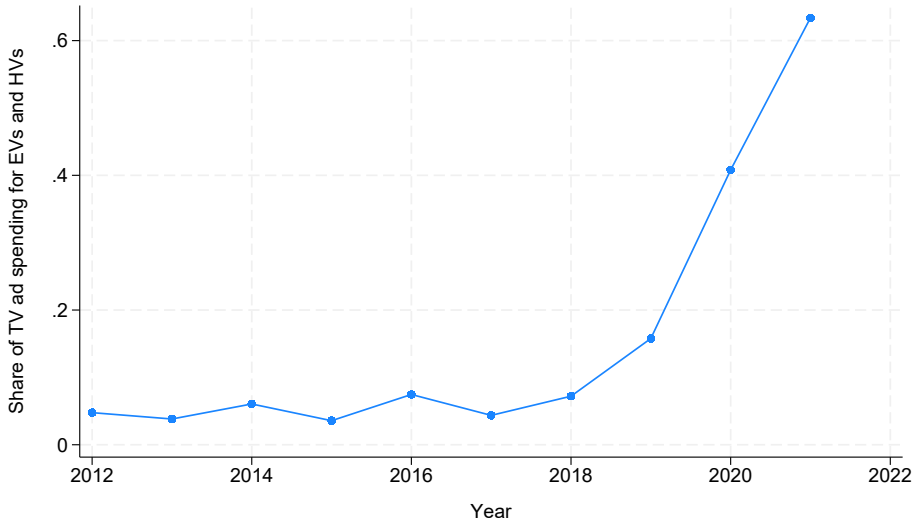


Figure A5: Expenditure Share of Electric and Hybrid Vehicles in TV Advertising

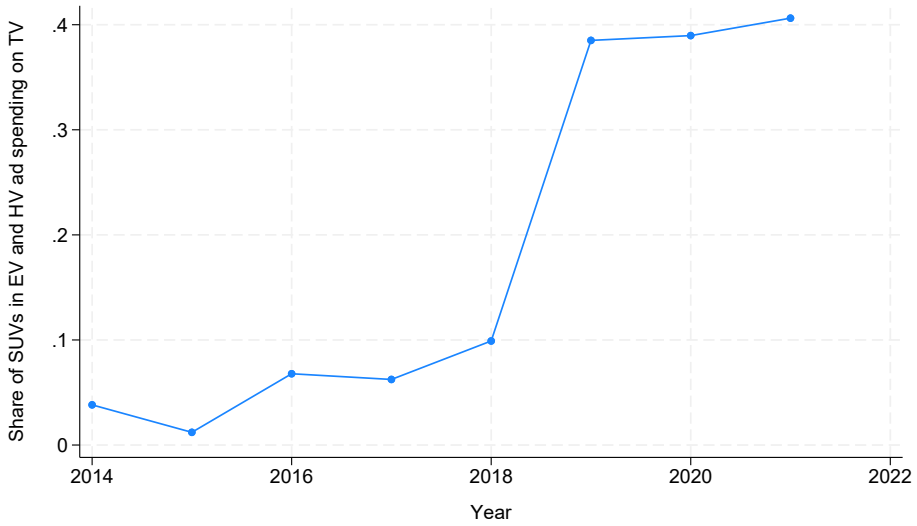


Figure A6: Expenditure Share of Electric and Hybrid SUVs in TV Advertising

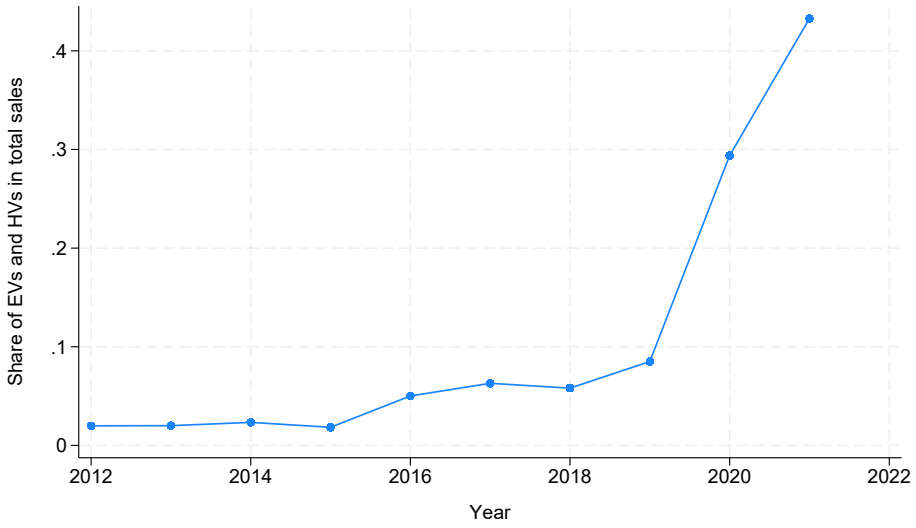


Figure A7: Share of Electric and Hybrid Vehicles in Total Sales

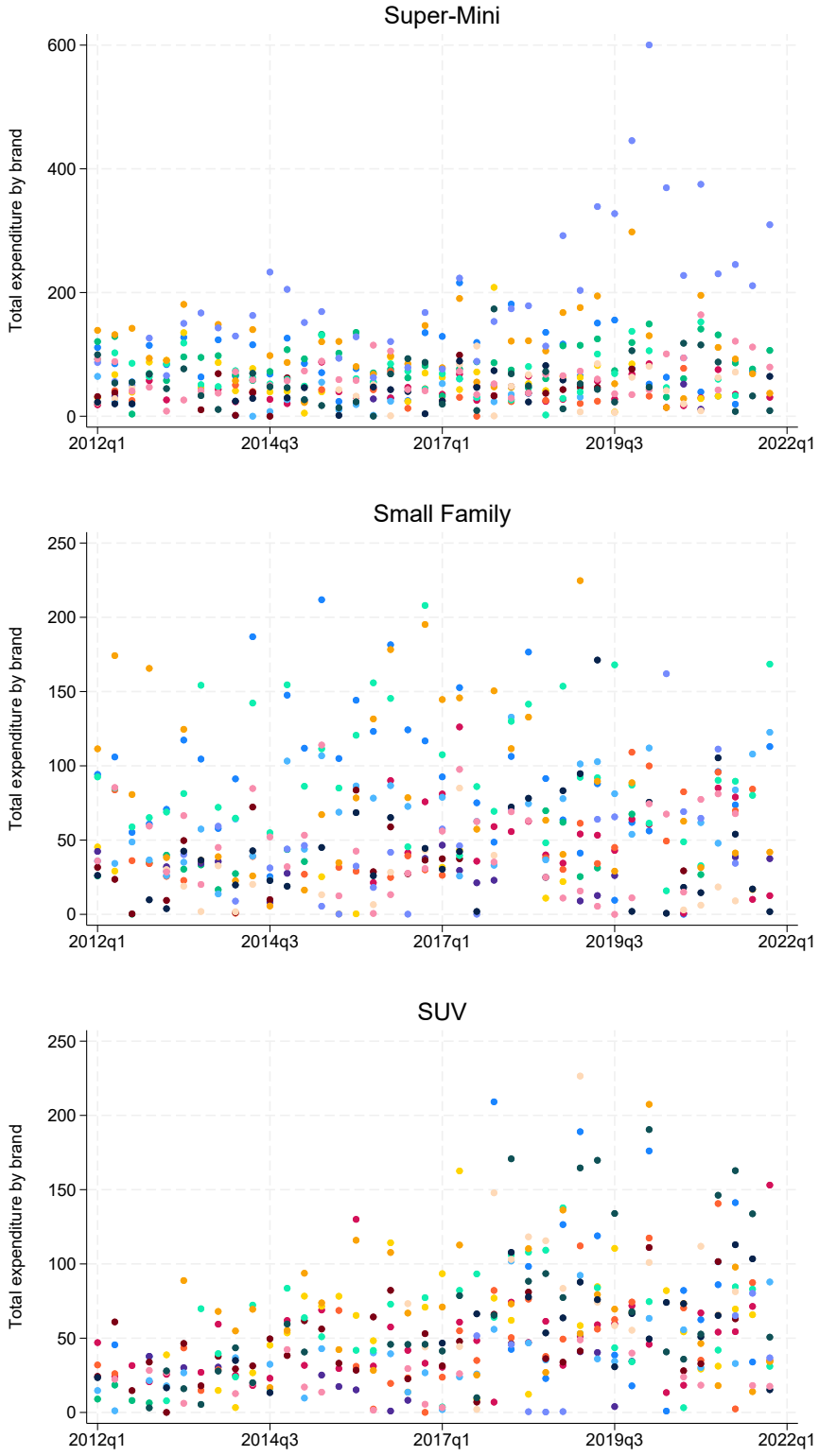


Figure A8: Total Advertising Expenditure by Brand and Segment

Note: Each dot represents a brand. Units of expenditures are not reported for confidentiality reasons.

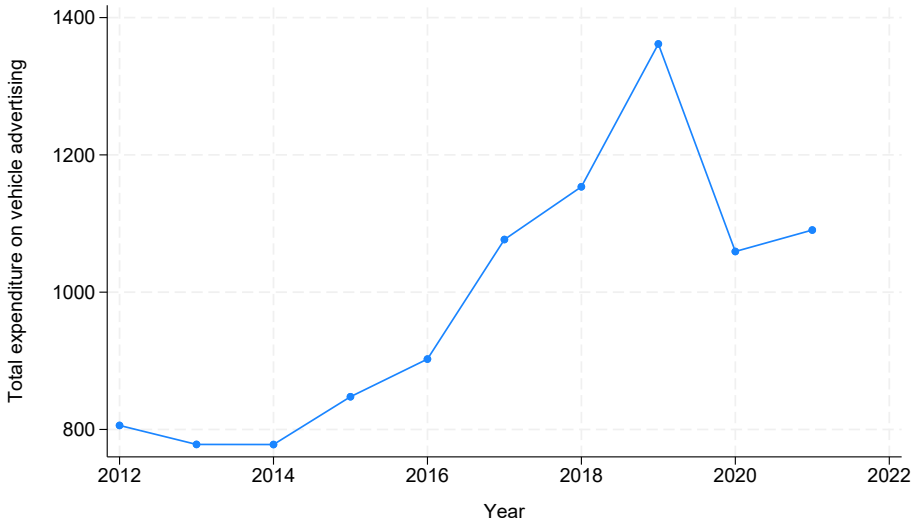


Figure A9: Total Expenditure on Vehicle Advertising

Note: Units of expenditures are not reported for confidentiality reasons.

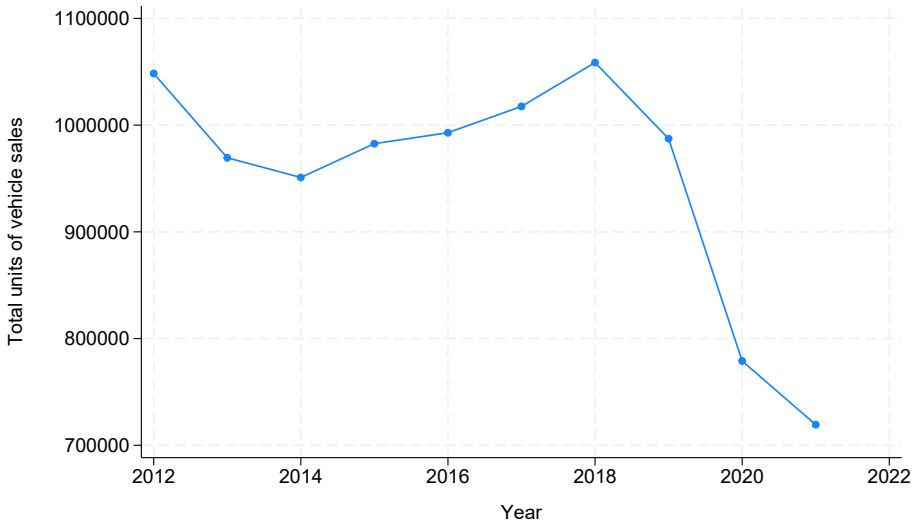


Figure A10: Total Units of Vehicle Sales

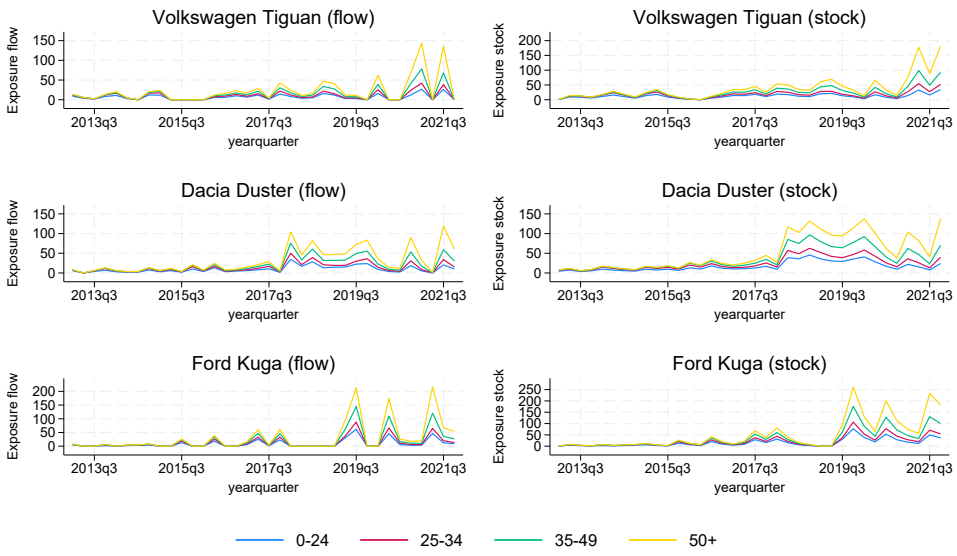


Figure A11: Flow and Stock of Advertising Exposure to Three SUV Models

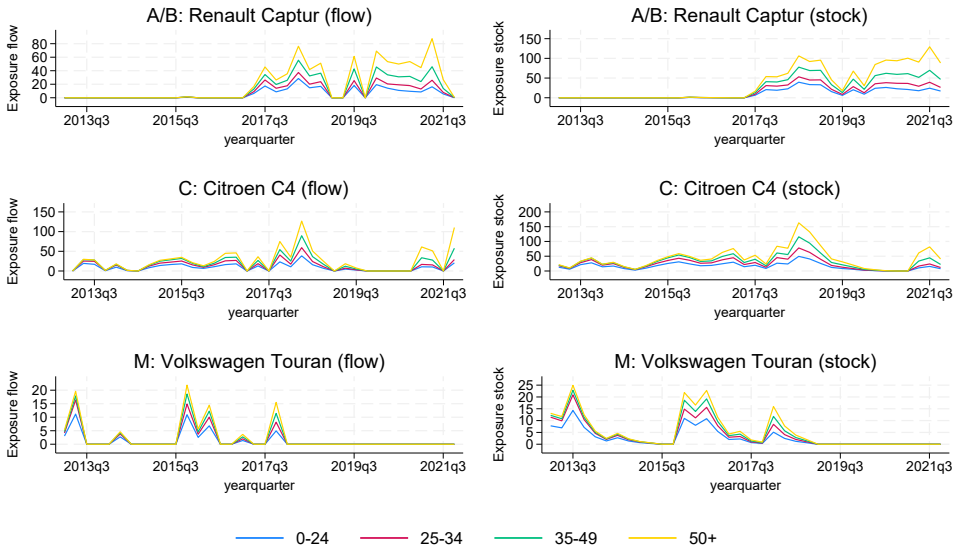


Figure A12: Flow and Stock of Advertising Exposure to Three Popular Models in Segments Mini (A/B), Small Family (C), and Minivan (M)

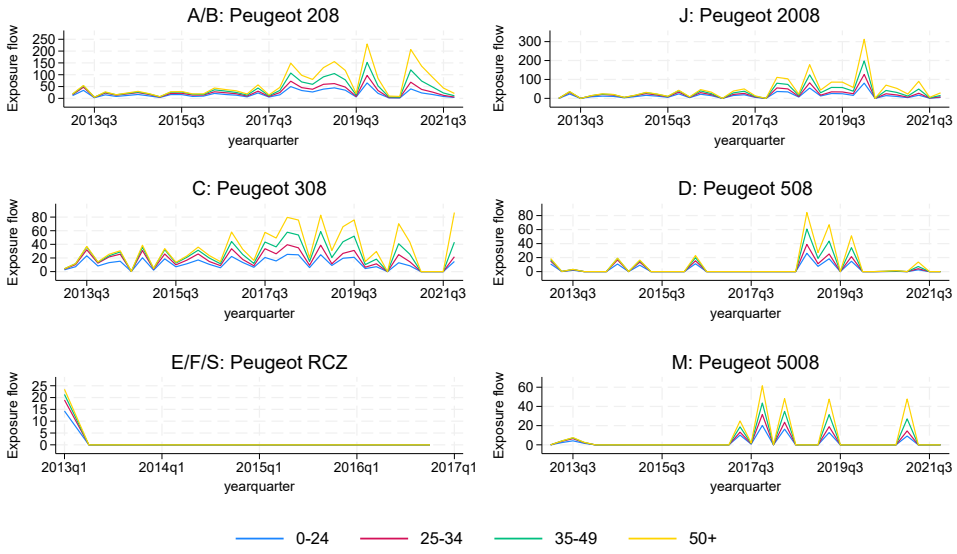


Figure A13: Flow of Advertising Exposure to Peugeot's Most Advertised Vehicles

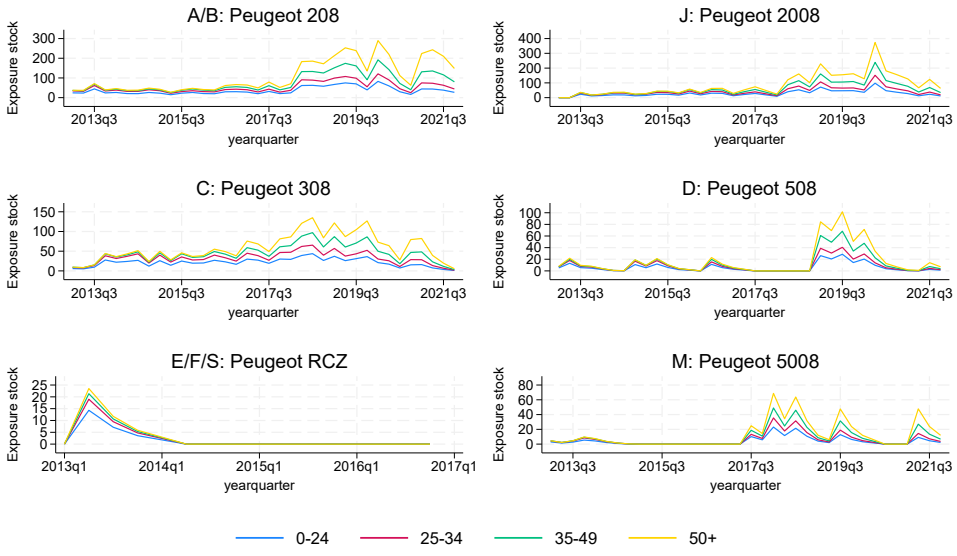


Figure A14: Stock of Advertising Exposure to Peugeot's Most Advertised Vehicles

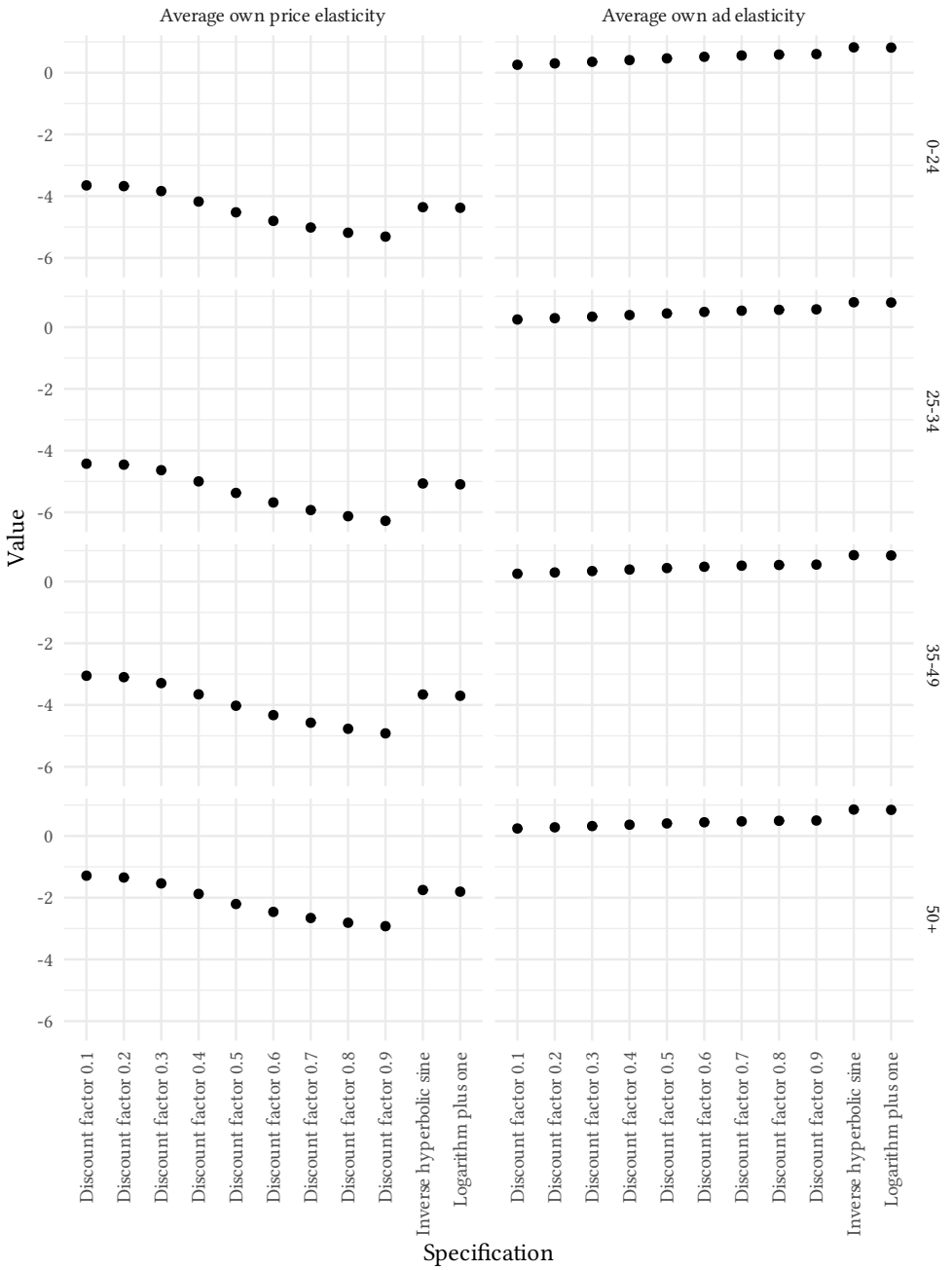


Figure A15: Robustness to Alternative Specifications of Advertising Stock

Note: Our baseline specification is defined in equation (2) with a discount factor of $\delta = 0.5$. Two alternative specifications that have also been used in the literature are the inverse-hyperbolic-sine and logarithm-plus-one specifications, both defined in footnote 10. In the figure, we plot the price and advertising elasticities obtained from equation (2) using nine alternative discount factors, together with the elasticities associated with the two alternative functional forms evaluated at $\delta = 0.5$.

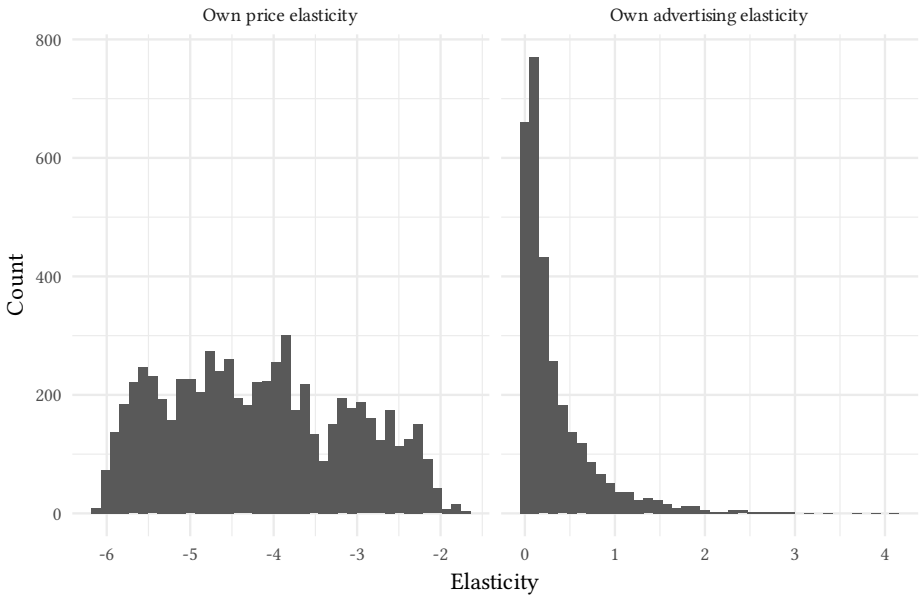


Figure A16: Distribution of Own Demand Elasticities

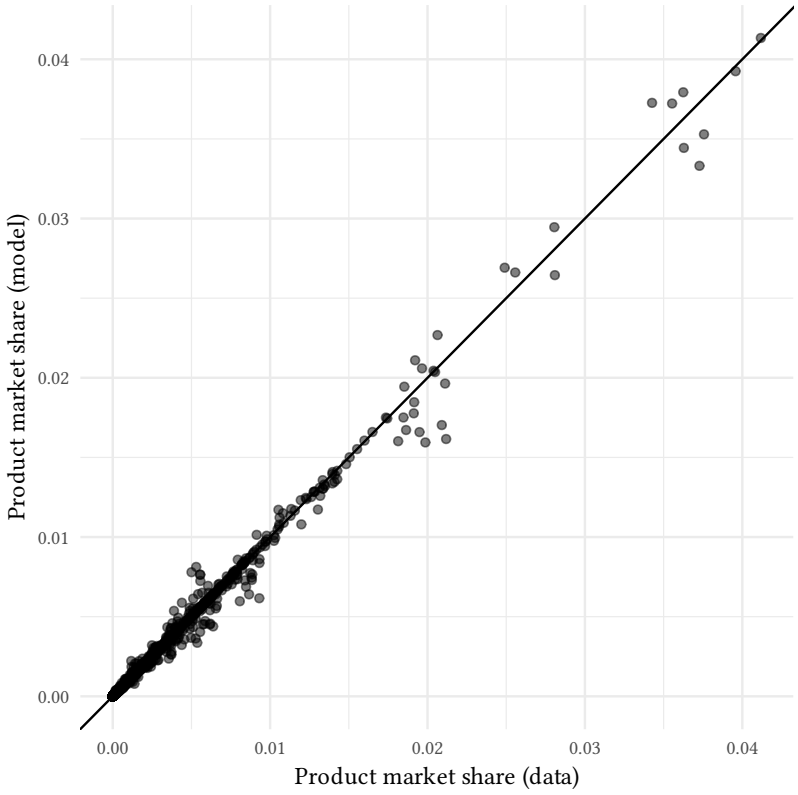


Figure A17: Model Fit: Predicted versus Observed Second-hand Sales, by Product

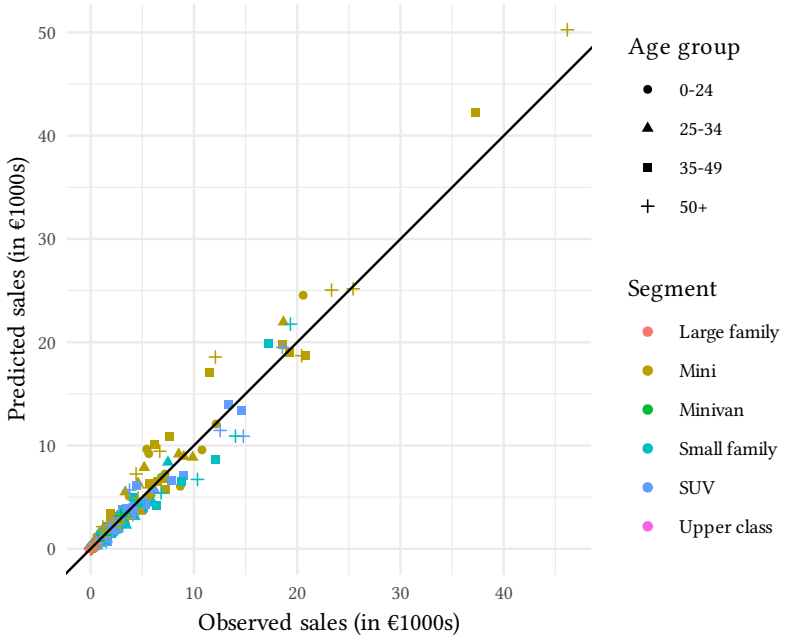


Figure A18: Model Fit: Predicted Sales versus Observed Sales, by Age-Segment-Brand

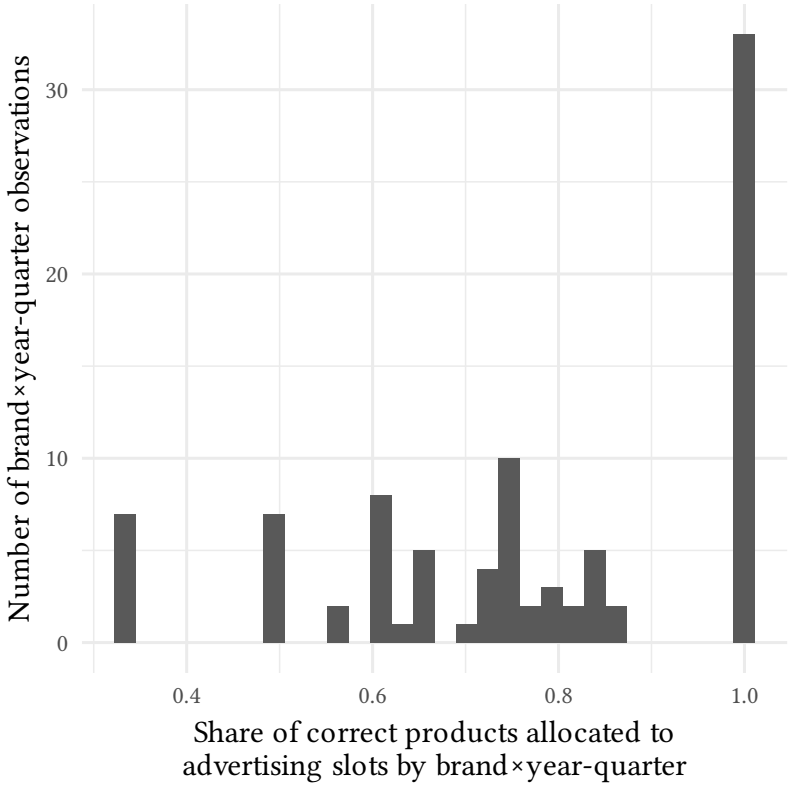


Figure A19: Model Fit: Share of correct products chosen to be advertised by brand-year-quarter. Excludes trivial cases where no or all products are advertised.

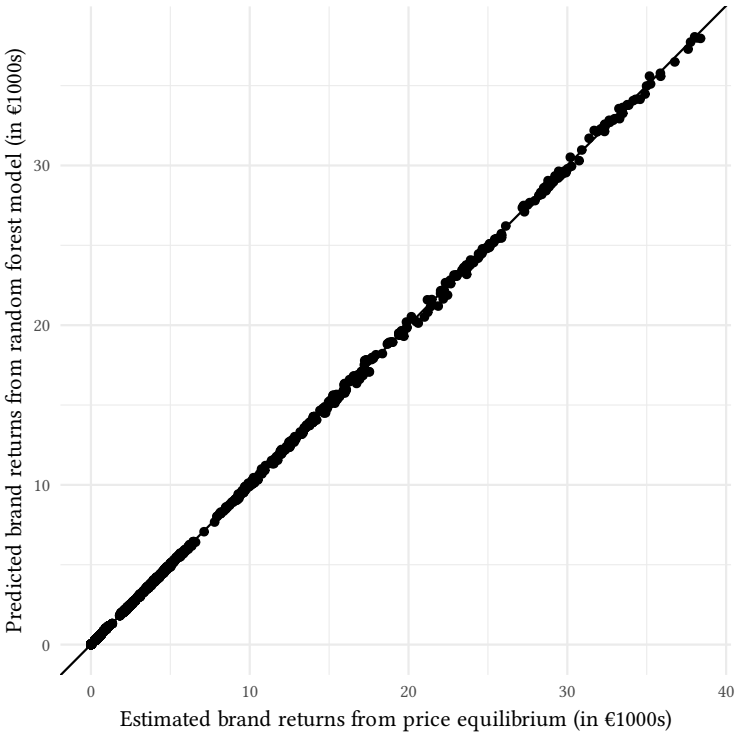


Figure A20: Out-of-Sample Fit of Random Forest Predictions for $\mathbb{E}[R_{b,t+1}]$

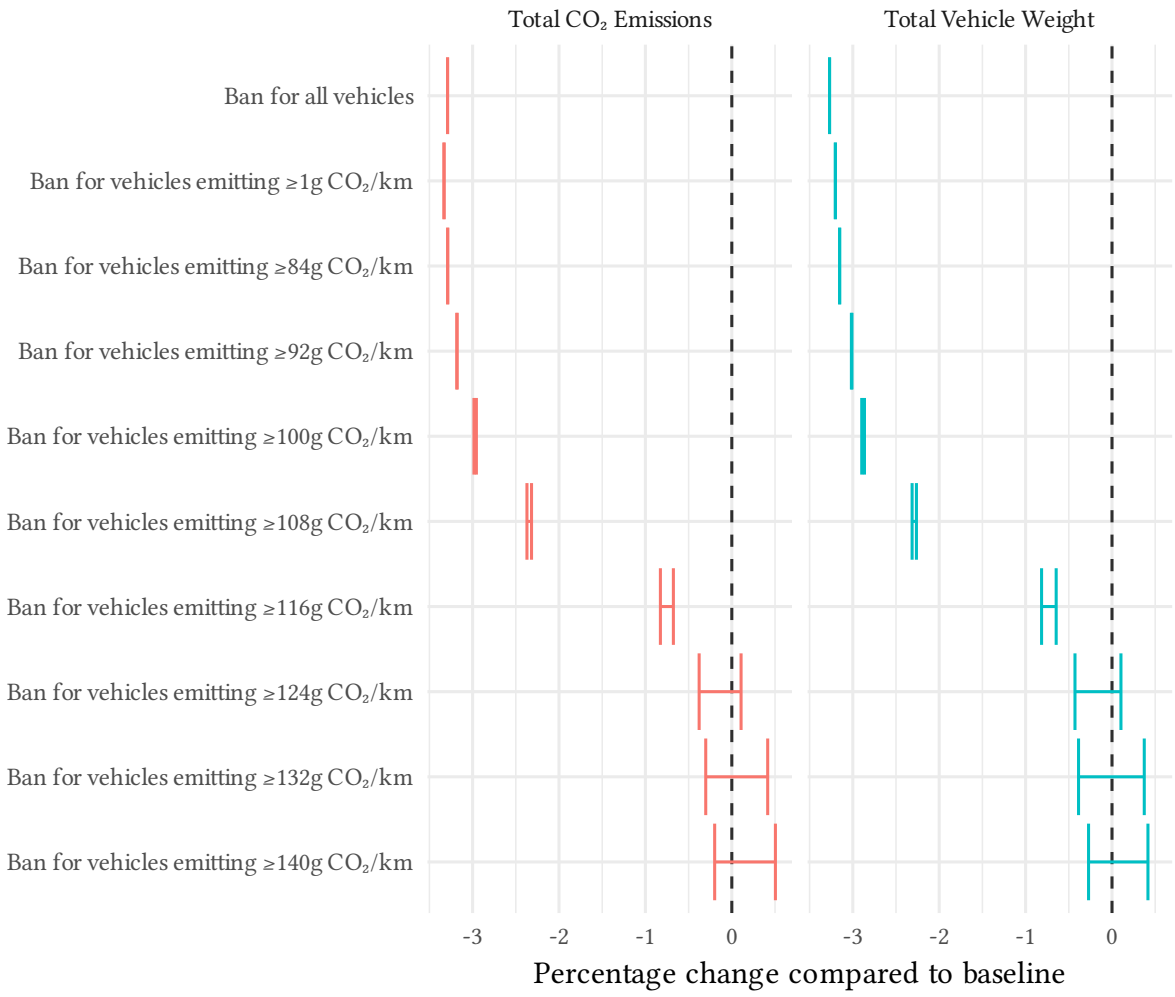


Figure A21: Environmental Effects of CO_2-Based Advertising Bans

Note: The whiskers indicate the range of environmental effects obtained across the simulated equilibrium bounds.

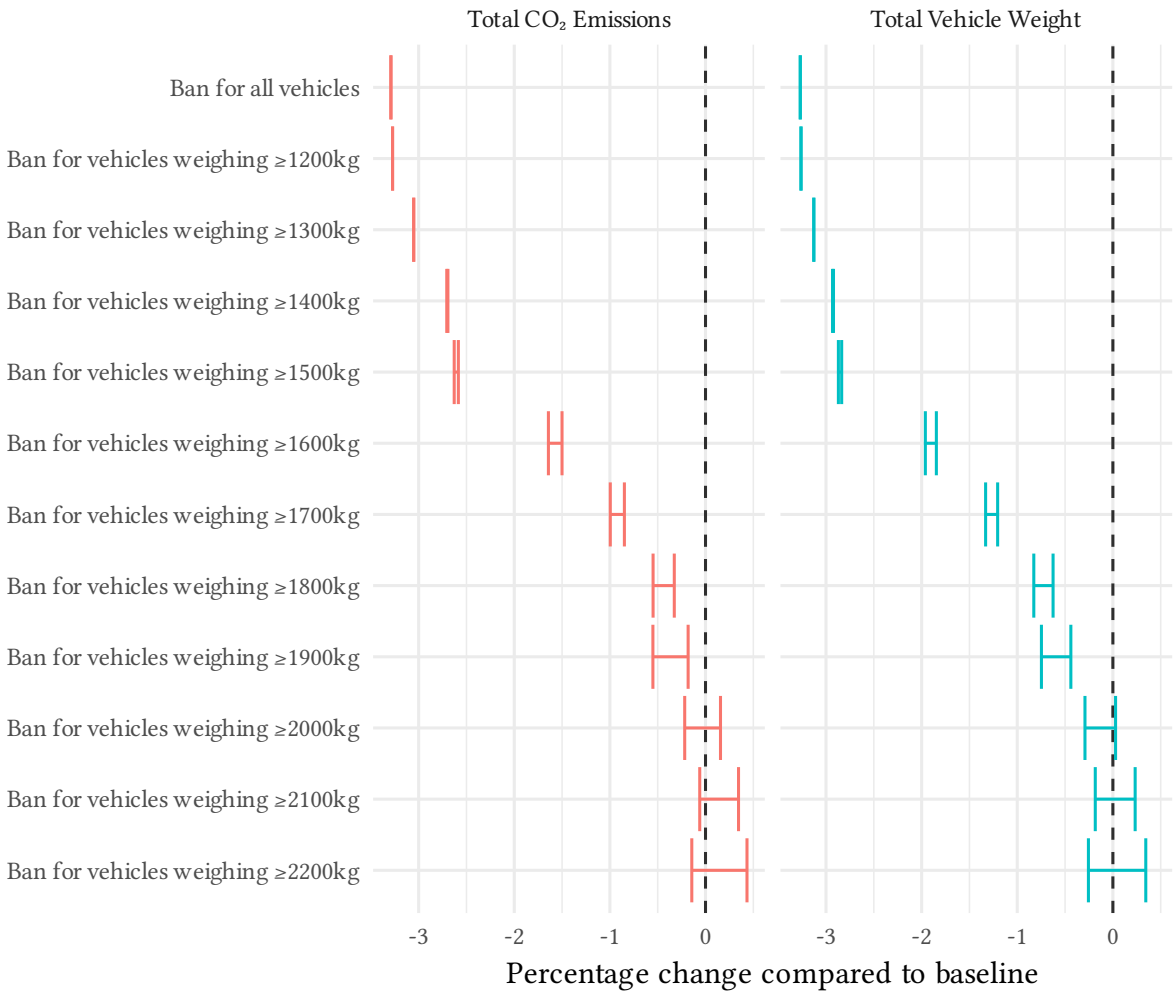


Figure A22: Environmental Effects of Weight-Based Advertising Bans

Note: The whiskers indicate the range of environmental effects obtained across the simulated equilibrium bounds.

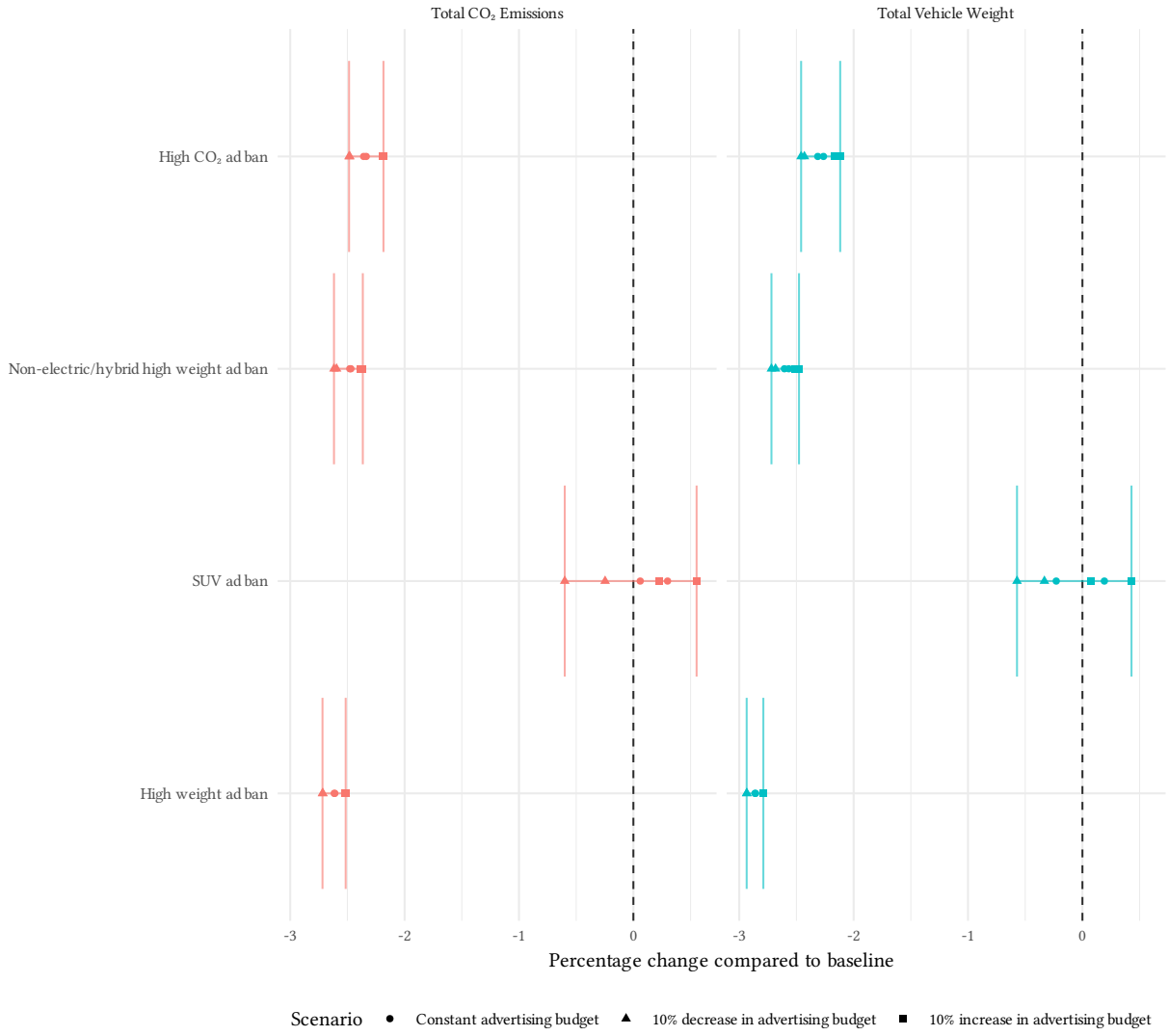


Figure A23: Robustness of Environmental Effects to Changes in Advertising Budgets

Note: Dots denote the equilibrium bounds under the baseline assumption of constant advertising budgets. Triangles denote the equilibrium bounds under a 10 percent reduction in vehicle manufacturers' advertising budgets, while squares denote the equilibrium bounds under a 10 percent increase in advertising budgets.

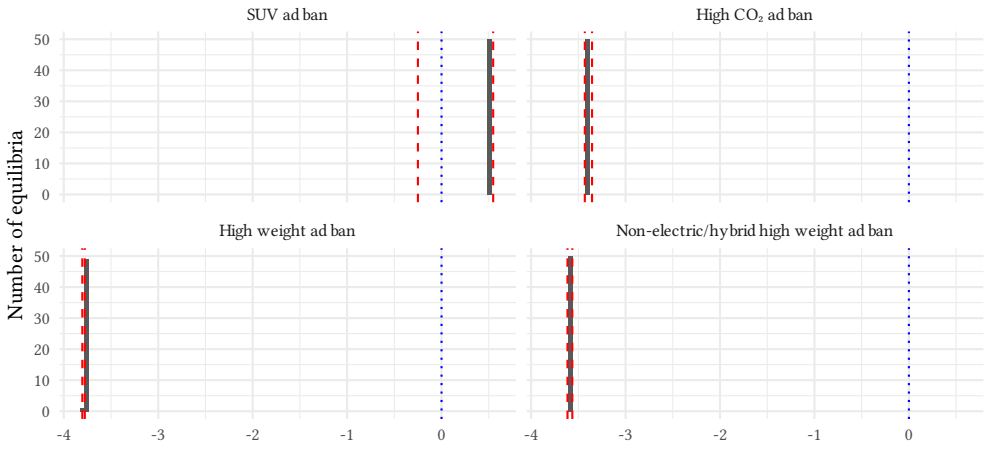


Figure A24: Equilibrium Percentage Changes in Total CO_2 Emission: Results from 50 Random Starting Allocations of Advertising Slots

Note: Dashed Lines Indicate the Equilibrium Bounds derived from the Extreme Starting Allocations

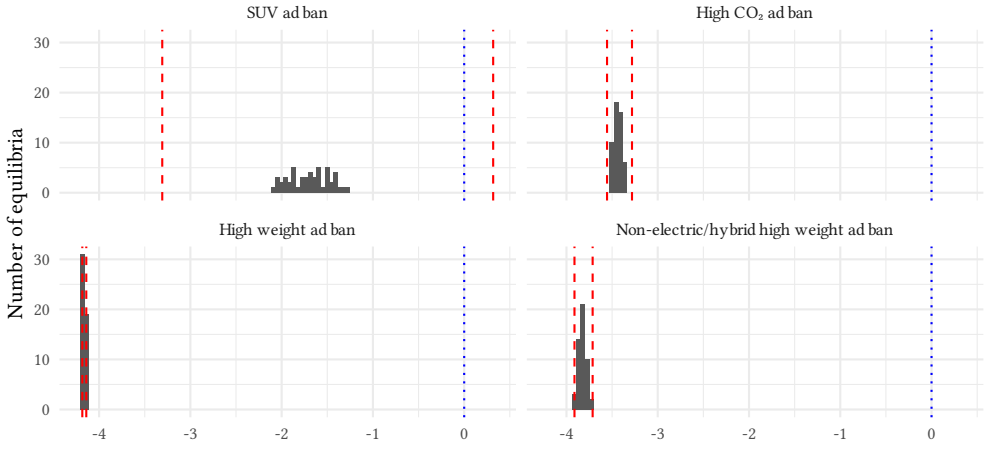


Figure A25: Equilibrium Percentage Changes in Total Vehicle Weight Level: Results from 50 Random Starting Allocations of Advertising Slots

Note: Dashed Lines Indicate the Equilibrium Bounds derived from the Extreme Starting Allocations

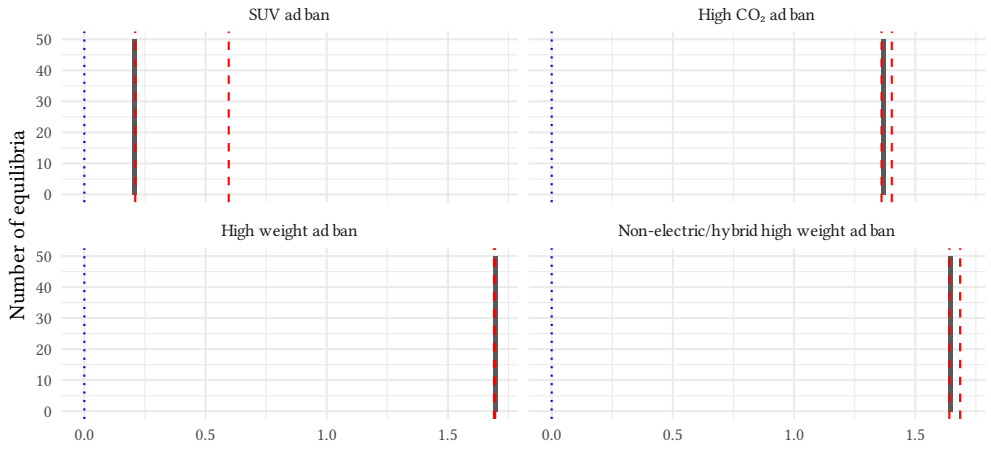


Figure A26: Equilibrium Changes in Consumer Surplus (in €bn): Results from 50 Random Starting Allocations of Advertising Slots

Note: Dashed Lines Indicate the Equilibrium Bounds derived from the Extreme Starting Allocations

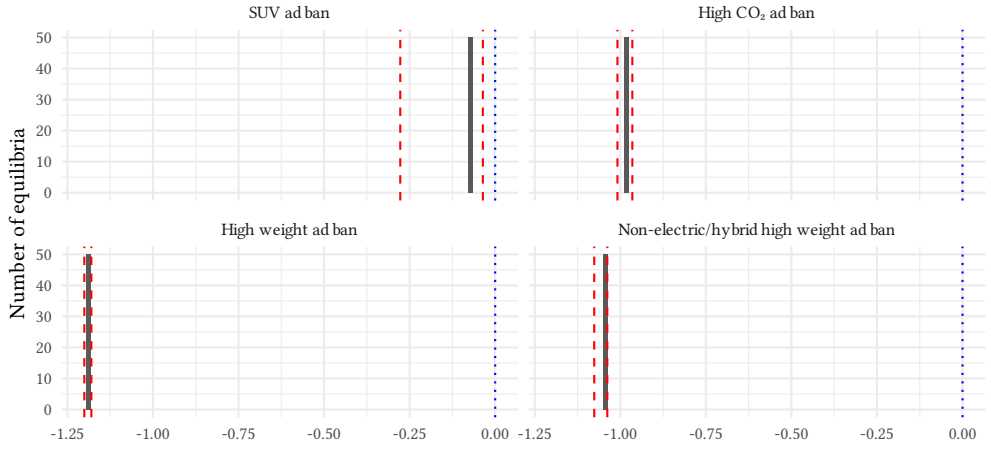


Figure A27: Equilibrium Changes in Total Firm Profit (in €bn): Results from 50 Random Starting Allocations of Advertising Slots

Note: Dashed Lines Indicate the Equilibrium Bounds derived from the Extreme Starting Allocations

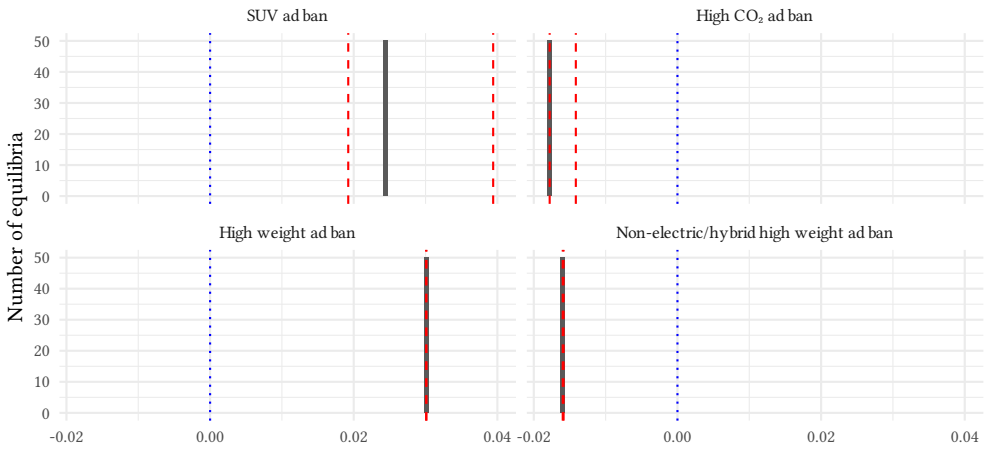


Figure A28: Equilibrium Changes in Total Feebate Revenue (in €bn): Results from 50 Random Starting Allocations of Advertising Slots

Note: Dashed Lines Indicate the Equilibrium Bounds derived from the Extreme Starting Allocations

Table A1: First-Stage Regression of Structural Demand Model

Dependent variable: Price	Coefficient	Standard Error
Included instruments:		
Ad exposure stock	0.040	(0.007)
Weight	132.908	(3.200)
Horsepower	108.346	(2.650)
Fuel cost	−4.230	(0.212)
IV Set 1: Rival vehicles' characteristics		
Constant	0.516	(0.094)
Weight	0.137	(0.046)
Horsepower	−0.243	(0.037)
Fuel cost	−0.075	(0.023)
IV Set 2: Rival vehicles' characteristics within segment		
Constant	−0.644	(0.129)
Weight	0.080	(0.119)
Horsepower	0.157	(0.118)
Fuel cost	0.647	(0.099)
IV Set 3: Brand's other vehicles' characteristics within segment		
Constant	2.767	(0.680)
Weight	−3.831	(1.159)
Horsepower	−5.194	(0.868)
Fuel cost	7.490	(0.703)
Vehicle body style fixed effects	Yes	
Age-product fixed effects	Yes	
Age-quarter-year fixed effects	Yes	
Segment-quarter-year fixed effects	Yes	
Observations	26384	

Note: Standard errors in parentheses.

Table A2: Average Product-Level Demand Elasticities

Age groups:	0-24	25-34	35-49	50+
Average own price elasticity	−4.521	−5.371	−4.023	−2.205
Average cross price elasticity	0.006	0.014	0.012	0.004
Average own ad elasticity	0.466	0.445	0.437	0.407
Average cross ad elasticity	−0.001	−0.002	−0.002	−0.002

Note: Elements in the table are calculated by taking the weighted average own- and cross-elasticities of vehicles, where each pair of vehicles is weighted by the product of their market shares.

Table A3: Average Cross-Elasticities by CO_2 Level

	CO_2 (g/km)	0-107	108-123	124+
Cross price elasticity				
	0-107	0.010	0.008	0.006
	108-123	0.010	0.008	0.006
	124+	0.011	0.009	0.007
Cross advertising elasticity				
	0-107	−0.002	−0.002	−0.001
	108-123	−0.002	−0.002	−0.001
	124+	−0.002	−0.002	−0.001

Note: Elements in the table are calculated by taking the weighted average cross-elasticities of vehicles within or across CO_2 levels, where each pair of vehicles is weighted by the product of their market shares. The table is to be read as the change in average demand for a vehicle of the emission level in a row with respect to a change in the average price (or advertising) of a vehicle of the emission level in a column.

Table A4: Average Cross-Elasticities by Weight Level

	Weight (kg)	0-1,499	1,500-2,199	2,200+
Cross price elasticity				
	0-1,499	0.003	0.008	0.003
	1,500-2,199	0.003	0.010	0.005
	2,200+	0.003	0.014	0.010
Cross advertising elasticity				
	0-1,499	0.000	−0.001	0.000
	1,500-2,199	−0.001	−0.002	0.000
	2,200+	−0.001	−0.003	−0.001

Note: Elements in the table are calculated by taking the weighted average cross-elasticities of vehicles within or across weight levels, where each pair of vehicles is weighted by the product of their market shares. The table is to be read as the change in average demand for a vehicle of the weight level in a row with respect to a change in the average price (or advertising) of a vehicle of the weight level in a column.

Table A5: Average Cross-Elasticities by Segment

Segment	Mini	Small family	Large family	Upper class	Minivan	SUV
Cross price elasticity						
Mini	0.008	0.008	0.002	0.002	0.002	0.007
Small family	0.010	0.011	0.004	0.003	0.003	0.010
Large family	0.012	0.015	0.006	0.005	0.005	0.013
Upper class	0.011	0.014	0.006	0.004	0.004	0.012
Minivan	0.010	0.011	0.004	0.003	0.003	0.009
SUV	0.010	0.011	0.004	0.004	0.003	0.010
Cross advertising elasticity						
Mini	−0.002	−0.001	0.000	0.000	0.000	−0.001
Small family	−0.002	−0.001	0.000	−0.001	0.000	−0.002
Large family	−0.003	−0.002	0.000	−0.001	0.000	−0.003
Upper class	−0.003	−0.002	0.000	0.000	0.000	−0.002
Minivan	−0.002	−0.001	0.000	−0.001	0.000	−0.002
SUV	−0.002	−0.002	0.000	−0.001	0.000	−0.002

Note: Upper class refers to vehicles in segment E/F/S (i.e., Executive/Luxury/Super vehicles). Elements in the table are calculated by taking the weighted average cross-elasticities of vehicles within or across segments, where each pair of vehicles is weighted by the product of their market shares. The table is to be read as the change in average demand for a vehicle of the segment in a row with respect to a change in the average price (or advertising) of a vehicle of the segment in column.

Table A6: Average Markup of Car Manufacturers

Firm	Avg. Markup	Firm	Avg. Markup
BMW	0.184	PSA	0.269
DAIMLER	0.208	RENAULT	0.366
FCA	0.285	RENAULT-NISSAN	0.261
FORD	0.261	RENAULT-NISSAN-MITSUBISHI	0.251
GEELY GROUP	0.181	STELLANTIS	0.243
GM	0.279	SUZUKI	0.353
HONDA	0.232	TATA	0.163
HYUNDAI	0.255	TOYOTA	0.235
MAZDA	0.215	VOLKSWAGEN	0.221
MITSUBISHI	0.275		

Note: The product-level markup is measured by the Lerner Index $L_{jt} = (\hat{p}_{jt} - c_{jt})/\hat{p}_{jt}$. The table presents the average product-level markup by firm.

Table A7: Robustness of Environmental Effects to the Keep-Old-Car Assumption

Percentage change in:	Total CO_2 emissions	Total vehicle weight
Outright ban	[-2.23, -2.23]	[-2.20,-2.20]
SUV ad ban	[-0.09, 0.28]	[-0.19,0.15]
High CO_2 ad ban	[-1.67, -1.64]	[-1.61,-1.58]
High weight ad ban	[-1.80, -1.78]	[-2.04,-2.03]
Non-electric/hybrid high weight ad ban	[-1.74, -1.71]	[-1.84,-1.81]

Note: All results [lower bound, upper bound] are percentage changes relative to the baseline equilibrium with no advertising ban. The high-CO_2 ad ban targets vehicles emitting more than 108g of CO_2 per km. The high-weight ad ban targets vehicles weighing more than 1,500 kg. The same weight threshold (1,500 kg) is applied in the non-electric/hybrid high-weight ad ban.

Table A8: Robustness of Welfare Effects to the Keep-Old-Car Assumption

	Change in consumer welfare	Change in firm profits	Change in feebate revenue	Change in social benefit	Change in total welfare
Outright ban	[2.11, 2.11]	[-1.62, -1.62]	[0.03, 0.03]	[0.03, 0.03]	[0.56, 0.56]
SUV ad ban	[0.21, 0.60]	[-0.28, -0.04]	[0.02, 0.04]	[-0.00, 0.00]	[0.16, 0.42]
High CO_2 ad ban	[1.36, 1.40]	[-1.01, -0.96]	[-0.02, -0.01]	[0.02, 0.02]	[0.39, 0.41]
High weight ad ban	[1.69, 1.70]	[-1.20, -1.18]	[0.03, 0.03]	[0.03, 0.03]	[0.55, 0.56]
Non-electric/hybrid high weight ad ban	[1.64, 1.69]	[-1.08, -1.04]	[-0.02, -0.02]	[0.02, 0.02]	[0.59, 0.62]

Note: All results [lower bound, upper bound] are in billions of euros.

B Price and Advertising Elasticities

B.1 Price Elasticities

The own-price elasticity is given by $\frac{\partial \mathfrak{s}_{jt}^d\left(\boldsymbol{\delta}_t^d,\sigma\right)}{\partial p_{jt}}\frac{p_{jt}}{\mathfrak{s}_{jt}^d}$, where:

$$\frac{\partial \mathfrak{s}_{jt}^d\left(\boldsymbol{\delta}_t^d,\sigma\right)}{\partial p_{jt}} = \int \alpha_i^d \widetilde{\mathfrak{s}}_{ijt}^d \left(1-\widetilde{\mathfrak{s}}_{ijt}^d\right)\phi\left(v_i\right)dv_i \tag{22}$$

The cross-price elasticity between products j and $j' \neq j$ is given by $\frac{\partial \mathfrak{s}_{jt}^d\left(\boldsymbol{\delta}_t^d,\sigma\right)}{\partial p_{j't}}\frac{p_{j't}}{\mathfrak{s}_{jt}^d}$, where

$$\left.\frac{\partial \mathfrak{s}_{jt}^d\left(\boldsymbol{\delta}_t,\sigma\right)}{\partial p_{j't}}\right|_{j,j'\neq j\in\mathcal{J}_t} = -\int \alpha_i^d \widetilde{\mathfrak{s}}_{ijt}^d \widetilde{\mathfrak{s}}_{ij't}^d \phi\left(v_i\right)dv_i \tag{23}$$

For notational simplicity, we have suppressed the arguments $\boldsymbol{\delta}_t^d, v_i$ and σ in $\widetilde{\mathfrak{s}}_{ijt}^d$ and $\widetilde{\mathfrak{s}}_{ij't}^d$. Both the own- and cross-price derivatives can be expressed compactly as:

$$\frac{\partial \mathfrak{s}_{jt}^d\left(\boldsymbol{\delta}_t,\sigma\right)}{\partial p_{j't}} = \int \alpha_i^d \left[-\widetilde{\mathfrak{s}}_{ijt}^d \widetilde{\mathfrak{s}}_{ij't}^d + \mathbb{1}\left\{j=j'\right\}\widetilde{\mathfrak{s}}_{ijt}^d\right]\phi\left(v_i\right)dv_i \tag{24}$$

where $\mathbb{1}\left\{\cdot\right\}$ is the indicator function.

B.2 Advertising Elasticities

The own-advertising elasticity is given by $\frac{\partial \mathfrak{s}_{jt}^d\left(\boldsymbol{\delta}_t^d,\sigma\right)}{\partial A_{jt}^d}\frac{A_{jt}^d}{\mathfrak{s}_{jt}^d}$, where:

$$\frac{\partial \mathfrak{s}_{jt}^d\left(\boldsymbol{\delta}_t,\sigma\right)}{\partial A_{jt}^d} = \int \beta^d \widetilde{\mathfrak{s}}_{ijt}^d \left(1-\widetilde{\mathfrak{s}}_{ijt}^d\right)\phi\left(v_i\right)dv_i \tag{25}$$

The cross-advertising elasticity is given by $\frac{\partial \mathfrak{s}_{jt}^d\left(\boldsymbol{\delta}_t^d,\sigma\right)}{\partial A_{j't}^d}\frac{A_{j't}^d}{\mathfrak{s}_{jt}^d}$, where

$$\left.\frac{\partial \mathfrak{s}_{jt}^d\left(\boldsymbol{\delta}_t,\sigma\right)}{\partial A_{j't}^d}\right|_{j,j'\neq j\in\mathcal{J}_t} = -\int \beta^d \widetilde{\mathfrak{s}}_{ijt}^d \widetilde{\mathfrak{s}}_{ij't}^d \phi\left(v_i\right)dv_i \tag{26}$$

Both the own- and cross-advertising derivatives can be expressed compactly as:

$$\frac{\partial \mathfrak{s}_{jt}^d\left(\boldsymbol{\delta}_t,\sigma\right)}{\partial A_{j't}^d} = \int \beta^d \left[-\widetilde{\mathfrak{s}}_{ijt}^d \widetilde{\mathfrak{s}}_{ij't}^d + \mathbb{1}\left\{j=j'\right\}\widetilde{\mathfrak{s}}_{ijt}^d\right]\phi\left(v_i\right)dv_i \tag{27}$$

C The Net Return to Advertising

To check whether the return to advertising from our demand estimates is consistent with the observed costs of advertising, we estimate the marginal returns of advertising at the observed advertising levels using our model estimates and compare them to our data on the marginal costs of advertising. Intuitively, while advertising boosts sales, its returns diminish as spending increases. Brands should continue advertising a product until its marginal return approaches its marginal cost and cease advertising once the marginal return falls below the marginal cost. For products where the marginal return of the first minute of advertising is below the marginal cost, brands will choose the corner solution of not advertising.

Formally, let $\widehat{T}_{jt} = \sum_{s \in \mathcal{S}_t} T_{jst}$ denote the total number of minutes of advertising across slots for product j in quarter t. The brand's marginal return in the following period to advertising today is given by:

$$\frac{\partial \mathbb{E}[R_{b,t+1}]}{\partial \widehat{T}_{jt}} = \mathbb{E}_{(\boldsymbol{\theta},\ \boldsymbol{A}_t^d,\ \mathcal{F}_{\xi_{t+1}^d},\forall d \in \mathcal{D})} \sum_{j' \in \mathcal{J}_{bt+1}} \left[\left(\hat{p}_{j',t+1}\left(\boldsymbol{A}_{t+1}\right) - c_{j',t+1} \right) \left(\sum_{d \in D} \frac{\partial \mathfrak{s}_{j',t+1}^d}{\partial A_{j,t+1}^d} \frac{\partial A_{j,t+1}^d}{\partial a_{jt}^d} \frac{\partial a_{jt}^d}{\partial \widehat{T}_{jt}} M_{t+1}^d \right) \right.$$
$$\left. + \frac{\partial \hat{p}_{j',t+1}\left(\boldsymbol{A}_{t+1}\right)}{\partial \widehat{T}_{jt}} \left(\sum_{d \in D} \mathfrak{s}_{j',t+1}^d M_{t+1}^d \right) \right]$$

where $\hat{p}_{j',t+1}\left(\boldsymbol{A}_{t+1}\right)$ denotes the equilibrium retail price of product j' in period $t+1$ given the vector of advertising stocks $\boldsymbol{A}_{t+1} = \left\{ \boldsymbol{A}_{t+1}^d \right\}_{d \in \mathcal{D}}$. The term $\frac{\partial \hat{p}_{j',t+1}(\boldsymbol{A}_{t+1})}{\partial \widehat{T}_{jt}}$ is the marginal change in the equilibrium price from a marginal increase in advertising.[53] Furthermore, we denote by κ_{bt} the marginal cost of advertising. For this we use the observed costs of the advertisements placed.

In equilibrium, we expect that for advertised vehicle-models the marginal benefit should approximately equal marginal cost:

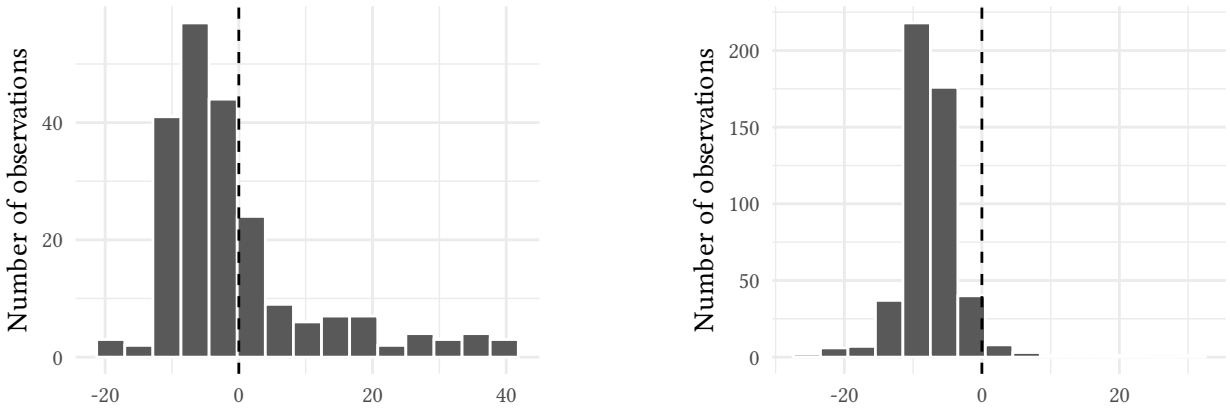
$$\frac{\partial \mathbb{E}[R_{b,t+1}]}{\partial \widehat{T}_{jt}} - \kappa_{bt} \simeq 0 \quad \forall j \in \mathcal{J}_t \text{ where } \widehat{T}_{jt} > 0$$

For models not advertised in equilibrium, we expect the marginal cost to exceed the marginal benefit for the first minute of advertising:

$$\frac{\partial \mathbb{E}[R_{b,t+1}]}{\partial \widehat{T}_{jt}} - \kappa_{bt} < 0 \quad \forall j \in \mathcal{J}_t \text{ where } \widehat{T}_{jt} = 0$$

Figure A29 plots the distributions of the model-predicted net returns on advertising, defined as

[53] Because the equilibrium price vector in period $t+1$ is an implicit function of the advertising minutes $\widehat{T}_{jt}$, in practice we compute the derivative $\frac{\partial \mathbb{E}[R_{b,t+1}]}{\partial \widehat{T}_{jt}}$ numerically. We first calculate the brand's expected return $\mathbb{E}[R_{b,t+1}]$ at the observed advertising levels. We then increase the observed number of advertising minutes for product j by a small amount ϵ, solve for the equilibrium price vector in period $t+1$, and recompute the brand's expected return. The numerical derivative is the change in brand's expected return divided by ϵ. In practice we use $\epsilon = 1 \times 10^{-9}$.



Advertised vehicle-quarters ($\widehat{T}_{jt} > 0$) Non-advertised vehicle-quarters ($\widehat{T}_{jt} = 0$)

Figure A29: Net Returns on Advertising (in thousands of euros)

$\frac{\partial \mathbb{E}[R_{b,t+1}]}{\partial \widehat{T}_{jt}} - \kappa_{bt}$, across our full estimation sample, split by observations with and without advertising. Consistent with brands' optimal advertising behavior, the estimated net returns are centered around zero for advertised vehicle-quarters ($\widehat{T}_{jt} > 0$), while they are predominantly negative for non-advertised ones ($\widehat{T}_{jt} = 0$).